The previous chapter showed an algebraic reformulation of quantum mechanics that only addressed the time-independent axiomatics. We ask how a quantum system evolves and how its continuous symmetries manifest. Instead of postulating a separate equation of motion or invoking group representations on a Hilbert space, we derive both the dynamics and the symmetry principles from the geometry as in Hamiltonian mechanics. The starting point is the set of pure states, which becomes a projective Hilbert space. This manifold carries a natural geometric structure that combines symplectic, Riemannian, and complex structures in a mutually compatible way called a Kähler structure.

The symplectic form provides the Poisson bracket that generates time evolution. Given the energy operator, its mean value becomes a smooth function on the state space, and the associated Hamiltonian vector field defines a flow whose integral curves are the solutions of the Schrödinger equation. The Riemannian metric captures the statistical and probabilistic aspects of the theory. In fact, it induces a notion of distance between pure states. Moreover, the geodesic separation between two neighbouring states equals the transition probability predicted by the Born rule. At the end, continuous symmetries of the system are implemented as flows generated by such admissible functions. Geometry guarantees the existence of a momentum map assigning to each symmetry generator a conserved quantity. This construction yields a geometric version of Noether's theorem. The entire structure thus follows from the intrinsic geometry of the state manifold, providing a fully geometrical formulation of quantum mechanics.

3.1 A Brief Introduction to Kähler Geometry

The mathematical framework used in this chapter is a generalization of the one used in Chapter **??**. We see that differential manifolds can be stated substituting open sets of $\mathbb{R}^n$ as the image of charts of the maximal atlas. In fact, to generalize differential geometry to complex numbers, we need an operation that mimics the complex unit, called a complex structure. Together with a particular metric and a manifold, this will create the key structure of the chapter, the Kähler manifold. This mathematical concept is rich enough to accommodate the whole framework of quantum mechanics, as we will see in the next sections.

We start by modeling a smooth manifold on a Hilbert space, as per Definition **??**. This statement follows Definition **??**, still we need to ensure compatibility with the structures already imposed on the Hilbert space.

Definition 3.1.1 (Hilbertian manifold)**.** *A real smooth Hilbertian manifold $\mathcal{M}$ modeled on a real Hilbert space $\mathcal{H}$, as per Definition **??**, is a smooth manifold, as per Definition **??**, whose maximal atlas $\mathfrak{A} = \{(U_\alpha, \phi_\alpha)\}_{\alpha \in I}$, as in Definition **??**, satisfies the following conditions.*

1. *For every $\alpha \in I$, the image $\phi_\alpha(U_\alpha)$ is an open subset of $\mathcal{H}$.*
2. *For every $\alpha, \beta \in I$ such that $U_\alpha \cap U_\beta \neq \emptyset$, the transition map $\phi_\beta \circ \phi_\alpha^{-1}$ is of class C^∞ in the sense of the Fréchet derivative.*

The almost complex structures and Riemannian metrics of an almost Kähler manifold act as sections of spaces of multilinear operators on the tangent space. Fixing notation for these spaces here avoids repeating it at every subsequent definition.

Definition 3.1.2 (Families of multilinear maps)**.** *Let $\mathcal{M}$ be a real smooth Hilbertian manifold, as per Definition 3.1.1. For each $x \in \mathcal{M}$, consider the Banach spaces of bounded linear operators on $T_x\mathcal{M}$, of bounded linear maps $T_x^*\mathcal{M} \to T_x\mathcal{M}$, and of bounded n-linear forms on $T_x\mathcal{M}$. The disjoint unions of these spaces over all $x \in \mathcal{M}$ are denoted, respectively, by*

$$\mathcal{B}(T\mathcal{M}) := \bigsqcup_{x\in\mathcal{M}} \mathcal{B}(T_x\mathcal{M}), \qquad \mathcal{B}(T^*\mathcal{M}, T\mathcal{M}) := \bigsqcup_{x\in\mathcal{M}} \mathcal{B}(T_x^*\mathcal{M}, T_x\mathcal{M}), \qquad \mathcal{B}_n(T\mathcal{M}, \mathbb{R}) := \bigsqcup_{x\in\mathcal{M}} \mathcal{B}_n(T_x\mathcal{M}, \mathbb{R}).$$

With this notation available, the almost complex structures needed to define an almost Kähler manifold can be introduced as smooth sections of $\mathcal{B}(T\mathcal{M})$ squaring to -1.

Definition 3.1.3 (Almost complex structure)**.** *Let $\mathcal{M}$ be a real smooth Hilbert manifold, as per Definition 3.1.1. An almost complex structure on $\mathcal{M}$ is a smooth section $J : \mathcal{M} \to \mathcal{B}(T\mathcal{M})$, as per Definition **??**, such that for every $x \in \mathcal{M}$, the image operator $J_x \in \mathcal{B}(T\mathcal{M})$ satisfies the property*

$$J_x \circ J_x = -\mathbb{1}_{T_x\mathcal{M}}.$$

Consequently, the map J endows each real tangent space with the algebraic structure of a complex vector space playing the role of the imaginary unit.

Example 3.1.4. On $\mathbb{R}^{2n} \cong \mathbb{C}^n$ with real coordinates $(x^1, \dots, x^n, y^1, \dots, y^n)$ and $z^j = x^j + iy^j$, the linear map defined pointwise by $J\partial_{x^j} = \partial_{y^j}$, $J\partial_{y^j} = -\partial_{x^j}$ satisfies $J^2 = -\mathbb{1}$. It descends that it is an almost complex structure as per Definition 3.1.3. It is the endomorphism induced by multiplication by i on $\mathbb{C}^n$ under the identification $T_z\mathbb{C}^n \cong \mathbb{C}^n$, and its integrability, in the sense of Definition 3.1.5 comes from the trivial holomorphic atlas of $\mathbb{C}^n$.

Not every almost complex structure comes from an underlying complex manifold structure. The obstruction is measured by a torsion tensor, and the case in which it vanishes deserves its own name because it is the case relevant to Kähler geometry.

Definition 3.1.5 (Integrable almost complex structure). *Let $\mathcal{M}$ be a real smooth Hilbert manifold, as per Definition 3.1.1. An almost complex structure J as per Definition 3.1.3 is integrable if its torsion vanishes. That is, if for every open set $\mathcal{U} \subset \mathcal{M}$ and every pair of smooth vector fields X, Y on $\mathcal{U}$,*

$$[JX, JY] - [X, Y] - J[X, JY] - J[JX, Y] = 0.$$

Having set the complex structure, the second needed ingredient is a metric.

Definition 3.1.6 (Riemannian metric). *Let $\mathcal{M}$ be a real smooth Hilbertian manifold, as per Definition 3.1.1. A Riemannian metric on $\mathcal{M}$ is a smooth section $g : \mathcal{M} \to \mathcal{B}_2(T\mathcal{M}, \mathbb{R})$, as in Definition 3.1.2, such that, for each $x \in \mathcal{M}$, the evaluation $g_x : T_x\mathcal{M} \times T_x\mathcal{M} \to \mathbb{R}$ is a symmetric, continuous, and positive-definite bilinear form.*

Remark 3.1.7 (Local coordinate expression and line element). In a local coordinate chart (U, x) with real coordinates $(x^1, \dots, x^n)$, the Riemannian metric g is a symmetric $(0, 2)$-tensor field, as in Definition **??**. It is expressed via the tensor product as

$$g = \sum_{i,j=1}^{n} g_{ij} dx^i \otimes dx^j, \tag{3.1}$$

where $g_{ij} := g\left(\frac{\partial}{\partial x^i}, \frac{\partial}{\partial x^j}\right)$ are the smooth metric components and $\{dx^i\}$ constitutes the dual basis of the cotangent space. In physics literature, the metric is also conventionally denoted by the infinitesimal line element

$$ds^2 = \sum_{i,j=1}^{n} g_{ij} dx^i dx^j. \tag{3.2}$$

In this notation, the juxtaposition $dx^i dx^j$ is a formal abbreviation for the symmetric tensor product $\frac{1}{2}(dx^i \otimes dx^j + dx^j \otimes dx^i)$. Consequently, expressions such as dx^2 denote the rank-two tensor $dx \otimes dx$. When evaluated on a tangent vector field, this tensor returns the square of the vector's spatial projection along the associated coordinate axis.

Having set all the structures needed, we are ready to enrich a Hilbertian manifold with an almost complex structure and a Riemannian metric. This generates an almost Kähler manifold.

Definition 3.1.8 (Almost Kähler manifold). *An almost Kähler manifold is a triple $(\mathcal{M}, J, g)$, where $\mathcal{M}$ is a real smooth Hilbertian manifold,as per Definition 3.1.1, J is an almost complex structure as per Definition 3.1.3, and g is a Riemann metric, as in Definition 3.1.6 such that*

(i) g is J-invariant, that is, $g_x(J_xX_x, J_xY_x) = g_x(X_x, Y_x)$ for all $x \in \mathcal{M}$, $X_x, Y_x \in T_x\mathcal{M}$.

(ii) The fundamental form defined via $\omega_x(X_x, Y_x) := g_x(J_xX_x, Y_x)$ is closed.

Remark 3.1.9. Definition 3.1.8 intertwines Riemannian, complex, and symplectic structures.

- Condition (i) makes J a metric isometry, yielding an almost Hermitian structure.

- Condition (ii) ensures the fundamental 2-form $\omega = g(J\cdot,\cdot)$ is closed and hence non-degenerate, elevating it to a symplectic form. Thus an almost Kähler manifold is a symplectic manifold, as in Definition **??**, with compatible J and metric g.

Physically, the triad (g, J, ω) unifies Hamiltonian mechanics, via ω, quantum state distances via g, and complex phase via J. The qualifier "almost" reflects that J need not be integrable, *i.e.*, its torsion tensor need not vanish.

Once the almost complex structure of an almost Kähler manifold is required to be integrable in the sense of Definition 3.1.5, the manifold acquires an underlying holomorphic structure and deserves the sharper name of Kähler manifold, which is the setting in which the projective Hilbert space of a quantum system will be placed.

Definition 3.1.10 (Kähler manifold)**.** *An almost Kähler manifold $(\mathcal{M}, J, g)$ as per Definition 3.1.8 is called a Kähler manifold if its almost complex structure J is integrable, as per Definition 3.1.5.*

The fundamental form ω and the metric g of an almost Kähler manifold each induce an isomorphism from the cotangent to the tangent space at every point, and it is these two musical isomorphisms, rather than ω and g themselves, that will enter into the definition of the Poisson and Riemann brackets.

Definition 3.1.11 (Musical isomorphisms of an almost Kähler manifold)**.** *Let $(\mathcal{M}, J, g)$ be an almost Kähler manifold as per Definition 3.1.8, with fundamental form ω. The musical isomorphisms $I_x, G_x : T_x^*\mathcal{M} \to T_x\mathcal{M}$, for $x \in \mathcal{M}$, are the topological-linear isomorphisms defined implicitly by*

$$\omega_x(I_x\alpha_x, X_x) = \langle \alpha_x, X_x \rangle, \qquad g_x(G_x\alpha_x, X_x) = \langle \alpha_x, X_x \rangle, \tag{3.3}$$

for $\alpha_x \in T_x^*\mathcal{M}$, $X_x \in T_x\mathcal{M}$. The corresponding smooth sections of $\mathcal{B}(T^*\mathcal{M}, T\mathcal{M})$ are denoted I, G.

Remark 3.1.12. The two musical isomorphisms of Definition 3.1.11 are related by $G = J \circ I$. Indeed, for $\alpha_x \in T_x^*\mathcal{M}$ and $X_x \in T_x\mathcal{M}$, the definitions of ω and I_x yield

$$g_x(J_x I_x \alpha_x, X_x) = \omega_x(I_x\alpha_x, X_x) = \langle \alpha_x, X_x \rangle = g_x(G_x\alpha_x, X_x), \tag{3.4}$$

and non-degeneracy of g_x entails that $J_x I_x \alpha_x = G_x \alpha_x$ for every α_x, that is, $G = J \circ I$.

With the musical isomorphisms in place, two bilinear brackets between smooth functions can be defined, one recovering the ordinary symplectic Poisson bracket and the other, depending only on the metric, providing the dispersion structure that will later encode the probabilistic content of quantum mechanics.

Definition 3.1.13 (Poisson and Riemann brackets)**.** *Let $(\mathcal{M}, J, g)$ be an almost Kähler manifold as per Definition 3.1.8, and let $f, l \in C^\infty(\mathcal{M}, \mathbb{R})$. The Poisson bracket $\{f, l\} \in C^\infty(\mathcal{M}, \mathbb{R})$ and the Riemann bracket $((f, l)) \in C^\infty(\mathcal{M}, \mathbb{R})$ are defined via*

$$\{f, l\} := \langle df, I\, dl \rangle, \qquad ((f, l)) := \langle df, G\, dl \rangle. \tag{3.5}$$

Where I, G are the musical isomorphisms of Definition 3.1.11, and $\langle \cdot, \cdot \rangle$ is the inner product between forms and vectors. By bilinearity, both extend to smooth complex-valued functions on $\mathcal{M}$.

Example 3.1.14. On $\mathbb{R}^2$, identified with $\mathbb{C}$ via $z = x + iy$, define a 2-form, as in Definition **??** via $\omega = dx \wedge dy$. In addition, define a Riemannian metric, as in Definition 3.1.6 as $g = dx^2 + dy^2$. Moreover, let J be a complex structure, as in Definition 3.1.3 via $J\partial_x = \partial_y$.

For the coordinate functions $x, y \in C^\infty(\mathbb{R}^2, \mathbb{R})$, the differentials are dx and dy. By Definition 3.1.11, $I\,dy = \partial_x$ and $G\,dy = \partial_y$. Hence $\{x, y\} = \langle dx, \partial_x \rangle = 1$ recovers the canonical Poisson bracket, while $((x, y)) = \langle dx, \partial_y \rangle = 0$, since dx and dy are g-orthogonal. The two coordinate functions are canonically conjugate but statistically independent.

Remark 3.1.15. The Poisson bracket of Definition 3.1.13 is the standard symplectic bracket associated with ω as in Definition **??**, while the Riemann bracket, depending only on g, is not a symplectic invariant. Using $G = J \circ I$, as per Remark 3.1.12, together with the J-invariance of ω and g,

$$\{f, l\} = \omega(Idf, Idl) = \omega(Gdf, Gdl), \qquad ((f, l)) = g(Gdf, Gdl) = g(Idf, Idl). \tag{3.6}$$

The Poisson and Riemann brackets combine into a single complex-valued bracket, and together with the ordinary pointwise product they generate two further bilinear products on smooth functions, the $\circ_\nu$ and $*_\nu$ products, that will be the geometric counterpart of the anticommutator and of the operator product once the manifold is specialized to $\mathbb{P}\mathcal{H}$.

Definition 3.1.16 (Kähler bracket, $\circ_\nu$ and $*_\nu$ products)**.** *Let $(\mathcal{M}, J, g)$ be an almost Kähler manifold as per Definition 3.1.8, and let $f, l \in C^\infty(\mathcal{M}, \mathbb{C})$. The Kähler bracket between f and l is*

$$\langle f, l\rangle := ((f, l)) + i\{f, l\}, \tag{3.7}$$

where $\{\cdot,\cdot\}$ and $((\cdot,\cdot))$ are the brackets of Definition 3.1.13. For $\nu \in \mathbb{R}$, the $\circ_\nu$ product and the $*_\nu$ product between f and l are the smooth complex functions

$$f \circ_\nu l := \tfrac{1}{2}\nu((f, l)) + fl, \qquad f *_\nu l := \tfrac{1}{2}\nu\langle f, l\rangle + fl. \tag{3.8}$$

Remark 3.1.17. Equipped with the Riemann bracket, or alternatively with the $\circ_\nu$ product, $C^\infty(\mathcal{M}, \mathbb{R})$, respectively $C^\infty(\mathcal{M}, \mathbb{C})$, is a real, respectively complex, commutative, nonassociative algebra. The Kähler bracket and the $*_\nu$ product provide two further complex, nonassociative algebra structures on $C^\infty(\mathcal{M}, \mathbb{C})$. With either of these two products and with complex conjugation as involution, $C^\infty(\mathcal{M}, \mathbb{C})$ is an involutive algebra, as per [1, p. 2892].

The four products introduced in Definition 3.1.16 are not independent. Each can be recovered from any other two, a fact that will be used repeatedly to move between the symplectic and the Jordan-like descriptions of the same algebraic structure.

Proposition 3.1.18. *Let $(\mathcal{M}, J, g)$ be an almost Kähler manifold as per Definition 3.1.8, and let $f, l \in C^\infty(\mathcal{M}, \mathbb{C})$. Then, with $\circ_\nu$, $*_\nu$ and $\{\cdot,\cdot\}$ as per Definitions 3.1.13 and 3.1.16,*

$$f *_\nu l = f \circ_\nu l + \tfrac{i}{2}\nu\{f, l\}, \tag{3.9}$$

$$f \circ_\nu l = \tfrac{1}{2}(f *_\nu l + l *_\nu f), \tag{3.10}$$

$$\{f, l\} = \tfrac{1}{i\nu}(f *_\nu l - l *_\nu f) \tag{3.11}$$

The proof of this statement can be found in [1, Equations (2.10)–(2.12)]. Among the smooth functions on an almost Kähler manifold, a distinguished class is identified by the requirement that its Hamiltonian flow preserve the metric, and not only the symplectic form. This is the class that will be identified with the observables of a quantum system.

Definition 3.1.19 (Kählerian function)**.** *Let $(\mathcal{M}, J, g)$ be an almost Kähler manifold as per Definition 3.1.8, let $f \in C^\infty(\mathcal{M}, \mathbb{R})$, and let $X_f := Idf$, with I as per Definition 3.1.11. The function f is Kählerian if $\mathcal{L}_{X_f} g = 0$, where $\mathcal{L}$ denotes the Lie derivative, as in Definition* **??**. *If $f \in C^\infty(\mathcal{M}, \mathbb{C})$, f is Kählerian if* Ref *and* Imf *are. The set of Kählerian functions is denoted $\mathcal{K}(\mathcal{M}, \mathbb{R})$, respectively $\mathcal{K}(\mathcal{M}, \mathbb{C})$.*

Remark 3.1.20. The vector field $X_f = Idf$ of Definition 3.1.19 is the Hamiltonian vector field of f, and the condition $\mathcal{L}_{X_f} g = 0$ states that the Hamiltonian flow of f preserves the metric g. Since ω is closed and $\iota_{X_f}\omega = df$ by Definition **??**, Cartan's formula, Equation (**??**), yields

$$\mathcal{L}_{X_f}\omega = d\iota_{X_f}\omega + \iota_{X_f} d\omega = d(df) = 0.$$

The flow automatically preserves the fundamental form. Moreover, from $\mathcal{L}_{X_f} g = 0$ it then follows that $\mathcal{L}_{X_f} J = 0$ as well, because J is determined by ω and g through Definition 3.1.8. Therefore, if f is Kählerian, its Hamiltonian flow preserves the whole Kähler structure.

This closure is what will let the observables of a quantum system, once identified with Kählerian functions, be combined without leaving the class of admissible dynamical generators.

Proposition 3.1.21. *$\mathcal{K}(\mathcal{M}, \mathbb{R})$ as in Definition 3.1.19, respectively $\mathcal{K}(\mathcal{M}, \mathbb{C})$, is a Lie subalgebra of $C^\infty(\mathcal{M}, \mathbb{R})$, respectively $C^\infty(\mathcal{M}, \mathbb{C})$, with respect to the Poisson bracket of Definition 3.1.13.*

This statement is proved in [1, p.2892]

Example 3.1.22 (Kählerian functions and the $*_\nu$ product). Continuing the flat space setup on $\mathbb{R}^2$ with $\omega = dx \wedge dy$ and $g = dx^2 + dy^2$. Consider the Hamiltonian function of the isotropic harmonic oscillator, $H(x, y) = \frac{1}{2}(x^2 + y^2)$. Its differential is $dH = xdx + ydy$. Applying the musical isomorphism I, the associated Hamiltonian vector field evaluates to $X_H = I\, dH = y\partial_x - x\partial_y$. Evaluating the Lie derivative of the metric tensor with respect to this vector field yields

$$\mathcal{L}_{X_H} g = \mathcal{L}_{y\partial_x - x\partial_y}(dx \otimes dx + dy \otimes dy) = 0. \tag{3.12}$$

Hence, H is a Kählerian function as per Definition 3.1.19.

Furthermore, the geometric framework algebraically composes functions via the Kähler bracket. Using the vanishing Riemann bracket $((x, y)) = 0$ and the canonical Poisson bracket $\{x, y\} = 1$ established previously, Definition 3.1.16 yields the exact evaluation of the $*_\nu$ product

$$x *_\nu y = xy + \frac{\nu}{2}\big(((x, y)) + i\{x, y\}\big) = xy + \frac{i\nu}{2}. \tag{3.13}$$

Fixing $\nu = \hbar$, this geometric product reconstructs the associative non-commutative algebraic product of the quantum position and momentum observables for a flat phase space.

Complex manifolds and the Dolbeault operators

The definitions and theorems of the preceding sections can appear abstract when viewed through the lens of pure linear algebra on the Hilbert space $\mathcal{H}$. To ground these concepts and show the way for explicit computations, it is essential to examine the local coordinate picture. The fundamental simplification offered by Kähler geometry is the reduction of a complex, multi-component geometric object to a single, real-valued scalar function called the Kähler potential. We begin by setting the definition of a complex manifold which is the extension of a real manifold to complex numbers.

Definition 3.1.23 (Complex manifold)**.** *A complex manifold of complex dimension n is a smooth manifold $\mathcal{M}$, as in Definition* **??**, *of real dimension $2n$ together with an atlas of charts $\varphi_\alpha : U_\alpha \to \mathbb{C}^n$, $\mathcal{M} = \bigcup_\alpha U_\alpha$ such that every transition map $\varphi_\beta \circ \varphi_\alpha^{-1}$, between open subsets of $\mathbb{C}^n$, is holomorphic. Such an atlas is called a holomorphic atlas for $\mathcal{M}$.*

A holomorphic atlas as per Definition 3.1.23 determines an integrable almost complex structure on $\mathcal{M}$, obtained by transporting multiplication by i on $\mathbb{C}^n$ through each chart. The converse, that every integrable almost complex structure arises this way, is the content of the Newlander-Nirenberg theorem, recalled here without proof.

Theorem 3.1 (Newlander-Nirenberg)**.** *Let $\mathcal{M}$ be a smooth manifold of real dimension $2n$ equipped with an integrable almost complex structure J, as per Definition 3.1.5. Then $\mathcal{M}$ admits a holomorphic atlas, as per Definition 3.1.23, inducing J.*

This theorem is proved in [2, Theorem 7.4]. Fix a complex manifold $\mathcal{M}$ of complex dimension n as per Definition 3.1.23, and a local holomorphic chart (U, φ) with complex coordinates $z = (z^1, \ldots, z^n) \in \mathbb{C}^n$, so that $z^j = x^j + iy^j$, where $(x^1, \ldots, x^n, y^1, \ldots, y^n)$ are the corresponding local real coordinates on U. The complexified cotangent space at each point of U then splits according to the holomorphic and antiholomorphic differentials, and this splitting is what allows differential forms to be assigned a bidegree.

Definition 3.1.24 (Forms of type (p, q))**.** *Let (U, φ) be a local holomorphic chart on a complex manifold $\mathcal{M}$, as per Definition 3.1.23, with complex coordinates $z = (z^1, \ldots, z^n)$. The subspaces $\Lambda^{1,0}$ and $\Lambda^{0,1}$ of the complexified cotangent space $T^*\mathcal{M} \otimes \mathbb{C}$ at each point of U are the spans of $\{dz^1, \ldots, dz^n\}$ and of $\{d\bar{z}^1, \ldots, d\bar{z}^n\}$, respectively. A differential form on U is of type (p, q) if, at every point, it lies in*

$$\Lambda^{p,q} := \underbrace{\Lambda^{1,0} \wedge \cdots \wedge \Lambda^{1,0}}_{p} \wedge \underbrace{\Lambda^{0,1} \wedge \cdots \wedge \Lambda^{0,1}}_{q}. \tag{3.14}$$

The decomposition of forms by type, as per Definition 3.1.24, is compatible with the exterior derivative, and this compatibility is what makes it possible to split d into a holomorphic and an antiholomorphic piece.

Definition 3.1.25 (Dolbeault operators). *Let (U,φ) be a local holomorphic chart on a complex manifold $\mathcal{M}$, as per Definition 3.1.23, with complex coordinates $z=(z^1,\dots,z^n)$. On $C^\infty(U,\mathbb{C})$, the Dolbeault operators $\partial, \bar\partial$ are the linear differential operators defined locally by*

$$\partial f := \sum_{j=1}^{n} \frac{\partial f}{\partial z^j} dz^j, \qquad \bar\partial f := \sum_{j=1}^{n} \frac{\partial f}{\partial \bar z^j} d\bar z^j, \tag{3.15}$$

extended to forms of type (p,q), as per Definition 3.1.24, by $d=\partial+\bar\partial$, with ∂ raising p by one and $\bar\partial$ raising q by one.

Remark 3.1.26. The nilpotency of the exterior derivative, $d^2=0$, together with the splitting $d=\partial+\bar\partial$ of Definition 3.1.25, yields $0=d^2=\partial^2+\bar\partial^2+(\partial\bar\partial+\bar\partial\partial)$. Since ∂^2, $\bar\partial^2$ and $\partial\bar\partial+\bar\partial\partial$ raise the bidegree (p,q) of a form to $(p+2,q)$, $(p,q+2)$ and $(p+1,q+1)$ respectively, these three subspaces intersect trivially, so each term vanishes separately:

$$\partial^2=0, \qquad \bar\partial^2=0, \qquad \partial\bar\partial+\bar\partial\partial=0. \tag{3.16}$$

A closed real $(1,1)$-form on a complex manifold, such as the fundamental form of a Kähler manifold, admits a local description in terms of the Dolbeault operators just introduced, which is the analytic fact underlying the notion of Kähler potential.

Theorem 3.2 (Local $\partial\bar\partial$-lemma). *Let ω be a real, closed differential form of type $(1,1)$, as per Definition 3.1.24, on a complex manifold $\mathcal{M}$, as per Definition 3.1.23. For every point $p\in\mathcal{M}$, there exists a contractible local holomorphic chart (U,z) containing p and a smooth real-valued function $K\in C^\infty(U,\mathbb{R})$ such that, on U,*

$$\omega = i\partial\bar\partial K \tag{3.17}$$

For the proof, we will refer the reader to [3, Theorem 4.17]. This theorem introduces a key function used to compute the symplectic form on a complex manifold.

Definition 3.1.27 (Kähler potential). *Let ω be a real, closed $(1,1)$-form on a complex manifold $\mathcal{M}$, as per Definition 3.1.23, and let (U,z) be a local holomorphic chart. A smooth function $K\in C^\infty(U,\mathbb{R})$ such that $\omega=i\partial\bar\partial K$ on U, as provided by Theorem 3.2, is called a local Kähler potential for ω on (U,z).*

Remark 3.1.28. The local Kähler potential of Definition 3.1.27 is not unique. In fact, if K and K' are both local Kähler potentials for ω on (U,z), then $\partial\bar\partial(K-K')=0$ on U, so that $K-K'$ coincides with the real part of a holomorphic function. Conversely, adding to K the real part of any holomorphic function on U produces another local Kähler potential for ω.

Once a local Kähler potential is available, the whole Riemannian metric tensor of a Kähler manifold as in Definition 3.1.10, restricted to a finite-dimensional holomorphic chart, is determined by the mixed second-order derivatives of that single scalar function.

Theorem 3.3 (Local coordinate expression of the metric). *Let $(\mathcal{M},J,g)$ be a Kähler manifold, as per Definition 3.1.10, of finite complex dimension n, and let K be a local Kähler potential, as per Definition 3.1.27, for the fundamental form ω on a chart (U,z). The components of the metric tensor g in the coordinates $(z^1,\dots,z^n)$ are given by*

$$g_{j\bar k} = \frac{\partial^2 K}{\partial z^j \partial \bar z^k}, \tag{3.18}$$

and the resulting Hermitian matrix $(g_{j\bar k})$ is positive definite everywhere on U.

This statement is not proved here. For further readings, see [2, p.58].

Remark 3.1.29. The positive definiteness recorded in Theorem 3.3, in the coordinates $(z^1,\dots,z^n)$, is equivalent to the requirement $g(X,X)>0$ for every nonzero tangent vector X, which is already part of g being a Riemannian metric on $\mathcal{M}$ as per Definition 3.1.8.

Local Expression of Curvature

The same local potential that determines the metric also determines its curvature. This geometrical object measures the local deviation from a flat Euclidean space. Since the metric components $g_{j\bar{k}}$ are defined by the second derivatives of the Kähler potential K, the local obstruction to flattening the metric relies on the second derivatives of $g_{j\bar{k}}$. Consequently, the Riemann curvature tensor is algebraically encoded within the fourth-order mixed partial derivatives of the single scalar function K.

Definition 3.1.30 (Riemann curvature via Kähler potential)**.** *Let $(\mathcal{M}, J, g)$ be a Kähler manifold, as in Definition 3.1.10 with local holomorphic coordinates $(z^1, \dots, z^n)$ and Kähler potential K, as in Definition 3.1.27. The components of the Riemann curvature tensor $R_{j\bar{k}m\bar{l}}$ evaluated on holomorphic planes are uniquely determined by the metric components $g_{j\bar{k}} = \partial_j \partial_{\bar{k}} K$.*

In a holomorphic chart centered at a point, the Kähler potential can always be chosen so that the metric components satisfy $g(\partial_j, \partial_k)|_0 = \delta_{jk}$ and all first derivatives vanish at the origin. This is equivalent to the freedom to add the real part of a holomorphic function to K. In such adapted coordinates, the curvature tensor depends on the fourth derivatives of K, with no lower-order interference.

At any local origin $z = 0$, where the first derivatives of the metric vanish, $\partial_m g_{j\bar{k}}|_0 = 0$, the curvature tensor components reduce to

$$R_{j\bar{k}m\bar{l}} := -\frac{\partial^2 g_{j\bar{k}}}{\partial z^m \partial \bar{z}^l} = -\frac{\partial^4 K}{\partial z^j \partial \bar{z}^k \partial z^m \partial \bar{z}^l}.$$

Contracting $R_{j\bar{k}m\bar{l}}$ with X and $\bar{X}$, normalized by $g(X, X)$, yields the holomorphic sectional curvature. That is the curvature along the complex curve tangent to X. This scalar will characterise $\mathbb{P}\mathcal{H}$ in Theorem 3.5 once shown constant.

Definition 3.1.31 (Holomorphic sectional curvature)**.** *Let $(\mathcal{M}, J, g)$ be a Kähler manifold, as in Definition 3.1.10 with Riemann curvature as in Definition 3.1.30. Given a tangent vector $X = \sum v^j \partial_{z^j}$, the* holomorphic sectional curvature $H(X)$ *is the normalized restriction of the curvature tensor along the complex line generated by X:*

$$H(X) := \frac{\sum R_{j\bar{k}m\bar{l}} v^j \bar{v}^k v^m \bar{v}^l}{\left(\sum g_{j\bar{k}} v^j \bar{v}^k\right)^2}. \tag{3.19}$$

Before curving the manifold, it is worth checking that the formula returns the answer expected of the flat model, which carries no curvature at all.

Example 3.1.32 (Flat space). On $\mathbb{C}^n$ with the flat Kähler potential $K(z, \bar{z}) = \|z\|^2$, the metric components are $g_{j\bar{k}} = \partial_j \partial_{\bar{k}} K = \delta_{jk}$, constant on all of $\mathbb{C}^n$. Every derivative of $g_{j\bar{k}}$ vanishes identically, so $R_{j\bar{k}m\bar{l}} = 0$ and $H(X) = 0$ for every tangent vector X, as expected of Euclidean space regarded as a Kähler manifold.

The local Kähler potential thus determines the entire geometry, metric and curvature alike. The non-vanishing terms at the fourth order of the Taylor expansion of K provide the exact fundamental obstruction to mapping the manifold globally to a Euclidean space. This algebraic reduction establishes the explicit starting point for evaluating the constant holomorphic sectional curvature of the projective Hilbert space in Proposition 3.3.8 below.

3.2 The Projective Hilbert Space as a Kähler Manifold

We now specialize the general Kähler framework to the manifold of physical interest, namely the projective Hilbert space. This space is obtained by quotienting the nonzero vectors of a complex Hilbert space by complex scaling. We show that the quotient is the pure state space of the algebra of compact operators. Equipped with a holomorphic atlas, it acquires a complex structure, while the underlying inner product naturally induces both a Riemannian metric and a compatible symplectic form. These two structures, together with the complex structure already present, form the Fubini–Study geometry, which is the canonical Kähler structure of the pure state space. Then, we derive their explicit local coordinate expressions. This is achieved through the introduction of a global Kähler potential on the chart, as we saw in the previous section. We start by constructing the projective Hilbert space via the space of unit norm representatives of a Hilbert space.

Definition 3.2.1 (Space of unit norm representatives)**.** *Let $\mathcal{H}$ be a Hilbert space, as in Definition* **??***. The space of unit norm representatives is*

$$\mathbb{S}\mathcal{H} := \{\Psi \in \mathcal{H} \mid (\Psi, \Psi) = 1\}.$$

To construct the state manifold, the Hilbert space is quotiented by an equivalence relation that identifies vectors differing only by a non-zero complex scalar. This eliminates the unobservable scale and phase.

Definition 3.2.2 (Projective Hilbert space)**.** *Let $\mathcal{H}$ be a Hilbert space. On $\mathcal{H}^* := \mathcal{H} \setminus \{0\}$ define $\Psi \sim \Phi$ if $\Phi = z\Psi$ for some $z \in \mathbb{C} \setminus \{0\}$. The* projective Hilbert space *is*

$$\mathbb{P}\mathcal{H} := \mathcal{H}^*/\sim,$$

with $[\Psi]$ the equivalence class of $\Psi \in \mathcal{H}^*$.

The quotient operation on the full Hilbert space restricts to a phase quotient on the unit sphere, as shown in the next proposition.

Proposition 3.2.3. *Let $\mathbb{S}\mathcal{H}$ be the space of unit norm representatives as in Definition 3.2.1. Restricting the identification in Definition 3.2.2, $\sim$, to $\mathbb{S}\mathcal{H}$ yields $[\Psi] \mapsto \tau(\Psi)$. That is a bijection $\mathbb{P}\mathcal{H} \to \mathbb{S}\mathcal{H}/U(1)$.*

This statement is proved in [4, Section 2.5]. To connect the geometric quotient space $\mathbb{P}\mathcal{H}$ to the operational framework of Chapter **??**, the pure state space of an arbitrary C^*-algebra must be mapped to projective Hilbert spaces. The GNS construction establishes that the total pure state space resolves into a disjoint union of sectors, demonstrating that studying a single $\mathbb{P}\mathcal{H}$ is sufficient to capture the geometry of any superselection sector.

Theorem 3.4 (Pure state decomposition)**.** *The pure state space $\mathcal{P}(\mathfrak{A})$ of a C^*-algebra $\mathfrak{A}$ as in Definition* **??** *is a disjoint union $\mathcal{P}(\mathfrak{A}) = \bigcup_\alpha \mathbb{P}\mathcal{H}_\alpha$, where $\mathcal{H}_\alpha$ is the Hilbert space of the irreducible GNS representation, Definition* **??***, of a state in $\mathbb{P}\mathcal{H}_\alpha$. States in the same $\mathbb{P}\mathcal{H}_\alpha$ are equivalent, states in different $\mathbb{P}\mathcal{H}_\alpha$ are inequivalent, and each inclusion $\mathbb{P}\mathcal{H}_\alpha \hookrightarrow \mathcal{P}(\mathfrak{A})$ is continuous.*

This decomposition is proved in [4, Theorem 2.5.4]. In the specific case where the observable algebra consists of the compact operators, the pure state space reduces to a single sector.

Proposition 3.2.4. *Let $\mathbb{P}\mathcal{H}$ be a projective Hilbert space as in Definition 3.2.2. Under the bijection specified in Proposition 3.2.3, $\mathbb{P}\mathcal{H}$ is identified with the pure state space of $\mathfrak{B}_0(\mathcal{H})$.*

The identification with the pure states of $\mathfrak{B}_0(\mathcal{H})$ follows from the correspondence between vector states and rank-one projections [4, Corollary 2.5.3]. Having established $\mathbb{P}\mathcal{H}$ as the physical state space, it must be rewritten as a Hilbertian manifold, as per Definition 3.1.1. This requires defining a maximal atlas of charts on a Hilbert space. The local charts are constructed by projecting a neighborhood of a state onto the orthogonal complement of its representative vector.

Definition 3.2.5 (Local chart of $\mathbb{P}\mathcal{H}$)**.** *Let $\mathbb{P}\mathcal{H}$ be a projective Hilbert space as in Definition 3.2.2. For $\psi \in \mathbb{P}\mathcal{H}$ and for $\Psi \in \mathbb{S}\mathcal{H}$ a representative, $N_\psi := \{\varphi \in \mathbb{P}\mathcal{H} \mid (\Psi, \Phi) \neq 0\}$ with Φ any representative of φ, set $F_\psi : N_\psi \to \mathcal{H}$ as*

$$F_\psi(\varphi) := \frac{\Phi}{(\Psi, \Phi)} - \Psi. \tag{3.20}$$

The structural properties of the map F_ψ guarantee that it provides a valid coordinate system modeled on the closed subspace $\Psi^\perp$.

Proposition 3.2.6. *Let $\mathbb{P}\mathcal{H}$ be a projective Hilbert space as in Definition 3.2.2. For $\psi \in \mathbb{P}\mathcal{H}$, let F_ψ be the map and N_ψ be the set defined in 3.2.5. We show that*

- F_ψ *is well defined,*
- $F_\psi(\psi) = 0$,
- *the image of* F_ψ *is* $\Psi^\perp$,
- $F_\psi : N_\psi \to \Psi^\perp$ *is a homeomorphism.*

Hence, $(N_\psi, F_\psi, \Psi^\perp)$ *is a local chart of* $\mathbb{P}\mathcal{H}$, *as in Definition* **??**.

Proof. $\Phi/(\Psi,\Phi)$ is invariant under $\Phi \mapsto z\Phi$, $z \neq 0$, so F_ψ is well defined, and $F_\psi(\psi) = \Psi/(\Psi,\Psi) - \Psi = 0$. Also F_ψ takes values in $\Psi^\perp$ because $(\Psi, F_\psi(\varphi)) = (\Psi,\Phi)/(\Psi,\Phi) - 1 = 0$. The map $\chi \mapsto \tau(\Psi + \chi)$, $\chi \in \Psi^\perp$, is a well-defined two-sided continuous inverse, since $\Psi + \chi \neq 0$ and both maps are built from continuous algebraic operations on the domain $(\Psi,\Phi) \neq 0$. □

Definition 3.2.5 can be used together with its properties in Proposition 3.2.6 to build an atlas on $\mathbb{P}\mathcal{H}$, to endow it with the structure of a Hilbertian manifold. This is made explicit in the next proposition.

Proposition 3.2.7 (Holomorphic atlas)**.** *The collection of charts* $\{(N_\psi, F_\psi, \Psi^\perp)\}_{[\Psi]\in\mathbb{P}\mathcal{H}}$ *endows* $\mathbb{P}\mathcal{H}$ *with the structure of a Hilbertian manifold, as in Definition 3.1.1.*

The proof of this proposition is brought by [4, Section 2.5]. The assignment of this differential structure induces a manifold topology on $\mathbb{P}\mathcal{H}$. To be physically consistent with the algebraic formulation, this manifold topology must coincide with the operational weak-* topology. That governs the convergence of expectation values.

Remark 3.2.8 (Operational versus geometric topologies). The weak-* topology on the state space, as per Definition **??**, is defined operationally. A sequence of states ω_α converges to ω if and only if $\omega_\alpha(A) \to \omega(A)$ for every observable $A \in \mathfrak{B}_0(\mathcal{H})$. This topology measures statistical indistinguishability. Conversely, the manifold topology induced by the holomorphic atlas of Proposition 3.2.7 measures geometric proximity, determined by the norm distance of representative vectors in the underlying Hilbert space. We will see in Section 3.4 that these two notions coincide.

Proposition 3.2.9. *The manifold topology on* $\mathbb{P}\mathcal{H}$ *coincides identically with the weak***-topologies induced by the dual space embeddings* $\mathbb{P}\mathcal{H} \subset \mathfrak{B}_0(\mathcal{H})^*$ *and* $\mathbb{P}\mathcal{H} \subset \mathcal{B}(\mathcal{H})^*$, *as per Definition* **??**.

The proof of this statement is provided in [4, Proposition 2.5.2].

Remark 3.2.10 (Physical consistency of the manifold structure). The topological equivalence established in Proposition 3.2.9 guarantees that smooth curves in the projective manifold correspond to continuous variations in the expectation values of all physical observables. Without this topological coincidence, the differential geometric structures introduced subsequently, would be operationally meaningless with respect to the algebraic state space.

Corollary 3.2.11. *The pure state space of* $\mathfrak{B}_0(\mathcal{H})$ *equipped with the weak** *topology is homeomorphic to* $\mathbb{P}\mathcal{H}$ *equipped with the manifold topology.*

Proof. Combine the bijection of Proposition 3.2.3 with the topological equivalence of Proposition 3.2.9. □

This formulation establishes the topological and differentiable equivalence between the algebraic pure state space and the geometric manifold $\mathbb{P}\mathcal{H}$. Proceeding from this, the canonical metric and symplectic structures are introduced in the subsequent section.

Explicit Kähler structure

Using Proposition 3.2.6, we make explicit the construction of the Kähler manifold onto the projective Hilbert space. Let $\psi \in \mathbb{P}\mathcal{H}$ and let $\Psi \in \mathbb{S}\mathcal{H}$ be a normalized representative, as shown in Proposition 3.2.3. By Proposition 3.2.6, the holomorphic tangent space $T_\psi\mathbb{P}\mathcal{H}$ is canonically isomorphic to the complex subspace $\Psi^\perp \subset \mathcal{H}$. To construct the Riemannian and symplectic structures, $T_\psi\mathbb{P}\mathcal{H}$ must be modeled as a real tangent space. This is achieved through the realification of $\Psi^\perp$.

The realification procedure restricts the field of scalars of $\Psi^\perp$ from $\mathbb{C}$ to $\mathbb{R}$. Consequently, the original complex scalar multiplication by the imaginary unit i is reinterpreted as a real-linear endomorphism on the realified space.

$$J : \Psi^\perp \to \Psi^\perp, \qquad JV := iV,$$

satisfying $J^2 = -\mathbb{1}$. This is an almost complex structure on $\mathbb{P}\mathcal{H}$ as per Definition 3.1.3, integrable as per Definition 3.1.5 since it is induced by the holomorphic atlas of Proposition 3.2.7.
The Hilbert space inner product $(\cdot,\cdot)$ restricts to a Hermitian inner product on $\Psi^\perp$. Its real and imaginary parts,

$$g_0(U,V) := \mathrm{Re}(U,V), \qquad \omega_0(U,V) := \mathrm{Im}(U,V), \tag{3.21}$$

define a Riemannian metric g_0 on $\mathbb{P}\mathcal{H}$, as per Definition 3.1.6, and a real bilinear form ω_0, with $(U,V) = g_0(U,V) + i\omega_0(U,V)$. Since scalar products are conjugate-linear in the first argument, $(iU,iV) = (U,V)$. It descends that $g_0(JU,JV) = g_0(U,V)$. That agrees with condition (i) of Definition 3.1.8. Evaluating instead on (U,JV),

$$(U,JV) = (U,iV) = i(U,V) = -\omega_0(U,V) + ig_0(U,V),$$

and comparing real parts with $(U,V) = g_0(U,V) + i\omega_0(U,V)$ yields the compatibility identity

$$\omega_0(U,V) = g_0(JU,V). \tag{3.22}$$

Closedness of ω_0, condition (ii) of Definition 3.1.8, is verified below through its local expression via a Kähler potential. Granting this, $(\mathbb{P}\mathcal{H}, J, g_0)$ is a Kähler manifold as per Definition 3.1.10.
This triple $(\mathbb{P}\mathcal{H}, J, g_0)$ constitutes the unscaled Fubini–Study structure of the projective Hilbert space. It is canonically induced by the linear inner product of $\mathcal{H}$ without introducing external parameters. However, g_0 and ω_0 are dimensionless mathematical objects. To correctly encode physical dynamics and quantum statistical dispersion, this canonical structure must be uniformly scaled by a real parameter.

Local coordinate expression via the Kähler potential

The function $\Phi \mapsto \|\Phi\|^2$ on $\mathcal{H}^*$ is the natural invariant of the quotient defining $\mathbb{P}\mathcal{H}$. We note that for $\lambda \in \mathbb{C}\setminus\{0\}$,

$$\ln\|\lambda\Phi\|^2 = \ln\lambda + \ln\bar{\lambda} + \ln\|\Phi\|^2.$$

In addition, $\partial\bar\partial$ annihilates the holomorphic term $\ln\lambda$ and the antiholomorphic term $\ln\bar\lambda$. It descends that

$$\partial\bar\partial \ln\|\lambda\Phi\|^2 = \partial\bar\partial \ln\|\Phi\|^2$$

. Hence $K_0 := \frac{1}{2}\ln\|\Phi\|^2$ descends to a local Kähler potential, as per Definition 3.1.27, unique up to the real part of a holomorphic function by Remark 3.1.28.

We apply Theorem 3.3 on the chart $N_\psi \simeq \Psi^\perp \simeq \mathbb{C}^n$ of Definition 3.2.5. A nearby state is represented by $\Phi = \Psi + \sum_{m=1}^n z^m e_m$, with $\{e_m\}$ an orthonormal basis of $\Psi^\perp$ and $z = (z^1,\dots,z^n) \in \mathbb{C}^n$. Since Ψ is normalized and orthogonal to each e_m,

$$\|\Phi\|^2 = 1 + \|z\|^2.$$

The local Kähler potential is therefore $K_0(z,\bar z) = \frac{1}{2}\ln\big(1+\|z\|^2\big)$, and

$$\frac{\partial K_0}{\partial z^j} = \frac{\bar z^j}{2(1+\|z\|^2)}, \qquad g_{0,j\bar k} = \frac{\partial^2 K_0}{\partial z^j \partial \bar z^k} = \frac{\delta_{jk}(1+\|z\|^2) - \bar z^j z^k}{2(1+\|z\|^2)^2}.$$

At $z=0$ this yields $g_{0,j\bar k} = \delta_{jk}/2$, matching g_0 of (3.21) evaluated on the orthonormal basis $\{e_m\}$

$$g_0(\partial_j,\partial_k) = 2\mathrm{Re}(g_{0,j\bar k}) = 2g_{0,j\bar k} = \delta_{j\bar k}.$$

The fundamental form is $\omega_0 = i\sum_{j,k} g_{0,j\bar k}\, dz^j \wedge d\bar z^k$, which is closed by Remark 3.1.26, completing the verification that $(\mathbb{P}\mathcal{H}, J, g_0)$ is Kähler.

3.3 Schrödinger Equation as an Hamiltonian Flow

The Kähler structure (J, g_0) of Section 3.2 is fixed only up to an overall positive rescaling of g_0. In fact, replacing g_0 by λg_0, for any $\lambda > 0$, again satisfies Definition 3.1.8, since J-invariance and closedness of the fundamental form survive positive rescaling. This section fixes λ by requiring the resulting Hamiltonian flow to reproduce the Schrödinger equation, and shows that the value obtained is the one already implicit, on the algebraic side, in the convention $\frac{1}{i\hbar}[A,B]$ fixed for the Lie–Jordan algebra of Definition **??**.
We begin by recalling the map $\pi : \mathcal{H} \setminus \{0\} \to \mathbb{P}\mathcal{H}$, $x \mapsto [x]$, the quotient map defining $\mathbb{P}\mathcal{H}$ as in Definition 3.2.2.

Definition 3.3.1 (Mean value function)**.** *Given a Hilbert space $\mathcal{H}$, as in Definition* **??***, and a bounded operator $A \in \mathcal{B}(\mathcal{H})$, the* mean value function *of A is*

$$\langle A\rangle : \mathbb{P}\mathcal{H} \to \mathbb{C}, \qquad \langle A\rangle([x]) := \frac{(x, Ax)}{\|x\|^2}.$$

If A in Definition 3.3.1 is selfadjoint, then $\langle A\rangle$ is real valued. By Stone's Theorem **??**, a selfadjoint operator $H \in \mathcal{B}(\mathcal{H})$ generates the one-parameter unitary group $U_t = e^{i\alpha t H}$ for some $\alpha \in \mathbb{R}$. The physical postulate of Schrödinger dynamics is that a state $[x] \in \mathbb{P}\mathcal{H}$ evolves along

$$\gamma(t) := [U_t x] = [e^{-itH/\hbar} x], \qquad t \in \mathbb{R}.$$

Following this physical statement, we determine the normalization λ of g_0 for which γ is the integral curve of the Hamiltonian vector field $X_{\langle H\rangle}$ of Definition 3.1.19, with respect to $\omega = \lambda\omega_0$.
Let $v \in T_{[x]}\mathbb{P}\mathcal{H}$ be a tangent vector. According to Proposition 3.2.6, v is uniquely represented by a vector $\eta \in x^\perp \subset \mathcal{H}$, where $x \in \mathbb{S}\mathcal{H}$ is a normalized representative of the state $[x]$.
To compute the differential $d\langle H\rangle_{[x]}(v)$, evaluate the directional derivative of the mean value function along a smooth curve $\pi : \mathbb{R} \to \mathcal{H} \setminus \{0\}$ generating the tangent vector η. Define the linear path $\pi(s) := x + s\eta$. Its projection to $\mathbb{P}\mathcal{H}$ passes through $[x]$ at $s = 0$ with velocity v.
The evaluation of the mean value function along this curve yields the real-valued function

$$\langle H\rangle([\pi(s)]) = \frac{(x + s\eta, H(x + s\eta))}{\|x + s\eta\|^2}.$$

Expanding the inner products in the numerator and denominator by sesquilinearity produces

$$\langle H\rangle([\pi(s)]) = \frac{(x, Hx) + s(\eta, Hx) + s(x, H\eta) + s^2(\eta, H\eta)}{(x,x) + s(\eta, x) + s(x, \eta) + s^2(\eta, \eta)}.$$

Imposing the normalization $(x, x) = 1$ and the orthogonality conditions $(\eta, x) = 0$ and $(x, \eta) = 0$, the denominator simplifies to $1 + s^2\|\eta\|^2$. The differential is the first-order derivative with respect to the parameter s evaluated at $s = 0$

$$d\langle H\rangle_{[x]}(v) = \frac{d}{ds}\left(\frac{(x, Hx) + s(\eta, Hx) + s(x, H\eta) + s^2(\eta, H\eta)}{1 + s^2\|\eta\|^2}\right)\Bigg|_{s=0}.$$

Applying the quotient rule, the derivative of the denominator evaluates to zero at $s = 0$, leaving only the first derivative of the numerator. The differential consequently isolates the linear coefficients

$$d\langle H\rangle_{[x]}(v) = (\eta, Hx) + (x, H\eta) = 2\mathrm{Re}(\eta, Hx).$$

Set $\eta_H := -\frac{i}{\hbar}Hx$. Since $\eta \in x^\perp$, only the component of η_H orthogonal to x contributes to the pairings below. Moreover, η_H may represent the candidate vector field at $[x]$ directly. Using $\omega = \lambda\omega_0$ and $\omega_0(U, V) = \mathrm{Im}(U, V)$, and conjugate-linearity of $(\cdot,\cdot)$ in the first argument,

$$\omega_{[x]}(\eta_H, \eta) = \lambda\,\mathrm{Im}\big(-\tfrac{i}{\hbar}Hx, \eta\big) = \frac{\lambda}{\hbar}\,\mathrm{Im}\big(i(Hx, \eta)\big) = \frac{\lambda}{\hbar}\,\mathrm{Re}(Hx, \eta) = \frac{\lambda}{\hbar}\,\mathrm{Re}(\eta, Hx).$$

Comparing with $d\langle H\rangle_{[x]}(v)=2\mathrm{Re}(\eta,Hx)$ shows that η_H represents $X_{\langle H\rangle}([x])$, that is, $\iota_{\eta_H}\omega=d\langle H\rangle$, when

$$\lambda=2\hbar. \tag{3.23}$$

Equation (3.23) fixes the free normalization of Section 3.2. From here on,

$$g:=2\hbar\, g_0, \qquad \omega:=2\hbar\,\omega_0, \tag{3.24}$$

denote the physical Fubini–Study metric and symplectic form of $\mathbb{P}\mathcal{H}$, with I,G the associated musical isomorphisms of Definition 3.1.11. By linearity of the potential in its normalization, the local Kähler potential and metric of Section 3.2 rescale to

$$K=\hbar\ln\big(1+\|z\|^2\big), \qquad g_{j\bar{k}}=2\hbar\, g_{0,j\bar{k}}=\hbar\,\frac{\delta_{jk}(1+\|z\|^2)-\bar{z}^j z^k}{(1+\|z\|^2)^2}.$$

This construction can be condensed in the following proposition.

Proposition 3.3.2 (Hamiltonian flow equals Schrödinger flow)**.** *Let $H\in\mathcal{B}(\mathcal{H})$ be selfadjoint. The Hamiltonian flow of $\langle H\rangle$ with respect to ω is*

$$\Phi_t([x])=[e^{-itH/\hbar}x], \qquad t\in\mathbb{R},$$

which satisfies the Schrödinger equation $i\hbar\frac{d}{dt}\big(e^{-itH/\hbar}x\big)=He^{-itH/\hbar}x$.

Remark 3.3.3. The value $\lambda=2\hbar$ in (3.24) is the only normalization compatible with the Stone evolution $e^{-itH/\hbar}$. If the flow had been stated as $e^{-itH/\hbar'}$ for some $\hbar'\neq\hbar$, the computation above would have fixed $\lambda=2\hbar'$ instead.

This same normalization is what identifies the geometric $*_\nu$ product at $\nu=\hbar$ with the associative product $AB=A\circ B+\frac{i\hbar}{2}\{A,B\}$ of the Lie–Jordan algebra of Definition **??**. The complete discussion of this correspondence will be presented in Corollary 3.5.5. Proposition 3.3.4 captures the Lie structure isomorphism.

Proposition 3.3.4 (Poisson bracket and commutator)**.** *For selfadjoint $A,B\in\mathcal{B}(\mathcal{H})$,*

$$\{\langle A\rangle,\langle B\rangle\}=\big\langle\{A,B\}\big\rangle=\big\langle\frac{1}{i\hbar}[A,B]\big\rangle.$$

Proof. By Definition 3.1.13 and Definition 3.1.19, $\{f,l\}=df(X_l)$, so, using the representative of $X_{\langle B\rangle}$ found above,

$$\{\langle A\rangle,\langle B\rangle\}([x])=d\langle A\rangle_{[x]}\big(X_{\langle B\rangle}([x])\big)=2\mathrm{Re}\Big(-\tfrac{i}{\hbar}Bx,Ax\Big)=-\tfrac{2}{\hbar}\mathrm{Im}(Bx,Ax).$$

Since A,B are selfadjoint,

$$(x,[A,B]x)=(Ax,Bx)-(Bx,Ax)=-2i\,\mathrm{Im}(Bx,Ax),$$

then it yields $\big\langle\frac{1}{i\hbar}[A,B]\big\rangle([x])=\frac{1}{i\hbar}(x,[A,B]x)=-\frac{2}{\hbar}\mathrm{Im}(Bx,Ax)$, which agrees with $\{\langle A\rangle,\langle B\rangle\}([x])$. □

Remark 3.3.5. The map $A\mapsto\langle A\rangle$ is thus a Lie algebra homomorphism from the selfadjoint part of $\mathcal{B}(\mathcal{H})$, with bracket $\frac{1}{i\hbar}[\cdot,\cdot]$, to $C^\infty(\mathbb{P}\mathcal{H},\mathbb{R})$, with Poisson bracket $\{\cdot,\cdot\}$ with no extra numerical factor once the normalization of Equation (3.23) is in place.

Geometry of the flow

The flow of Proposition 3.3.2 is generated by the unitaries $U_t=e^{-itH/\hbar}$. Since each unitary preserves the Hilbert space inner product, its projectivization preserves g and ω, so each Φ_t is a Kähler isomorphism of $(\mathbb{P}\mathcal{H},\omega,g)$.

Proposition 3.3.6 (Kähler isomorphisms and their generators)**.** *Let $\mathcal{H}$ be a Hilbert space as in Definition **??**, and $\mathbb{P}\mathcal{H}$ be its projective space as in Definition 3.2.2. We can state the following.*

1. *Unitary operators on $\mathcal{H}$ induce Kähler isomorphisms of $\mathbb{P}\mathcal{H}$, and conversely every Kähler isomorphism is of this form, uniquely up to phase.*

2. *Selfadjoint operators generate one-parameter groups of Kähler isomorphisms via $\Phi_t([x]) = [e^{-itH/\hbar}x]$. Conversely every such group is generated by a unique H up to additive constant.*

Proof. The first statement follows because unitaries preserve the inner product, hence their projectivization preserves both ω and g. In addition, any Kähler automorphism of $\mathbb{P}\mathcal{H}$ lifts to a unitary or antiunitary operator on $\mathcal{H}$ by Wigner's Theorem, [5, Theorem 12.11], and restricting to continuous paths through the identity selects the unitary lifts.

For the second statement, $e^{-itH/\hbar}$ is a strongly continuous one-parameter unitary group by Stone's Theorem **??**, so its projectivization is a continuous group of Kähler isomorphisms. Conversely, any such group lifts to a strongly continuous one-parameter unitary group U_t, up to phases, whose selfadjoint generator H satisfies $U_t = e^{-itH/\hbar}$ by Stone's Theorem **??**, unique up to an additive real constant, which does not affect the projectivized flow. □

Remark 3.3.7. A quantum system is thus a Hamiltonian system. In particular, it is a symplectic manifold $(\mathbb{P}\mathcal{H}, \omega)$ together with a distinguished one-parameter group of symplectic, indeed Kähler, isomorphisms generated by the Hamiltonian function $\langle H \rangle$.

Curvature of the State Manifold

We now specialize the holomorphic sectional curvature of Definition 3.1.31 to the projective Hilbert space. A direct computation will show that in the case of $\mathbb{P}\mathcal{H}$ it is constant and positive. The general fact is stated by Hawley-Igusa Theorem 3.5 showing that projective spaces are the only possible choice for quantum mechanics.

Proposition 3.3.8 (Curvature of the projective state space)**.** *Let $\mathbb{P}\mathcal{H}$ be the projective Hilbert space, as in Definition 3.2.2, endowed with the Fubini–Study metric g_λ generated by the scaled local Kähler potential, as in Definition 3.1.27, via*

$$K_\lambda(z, \bar{z}) := \frac{\lambda}{2} \ln\left(1 + \|z\|^2\right),$$

where $\lambda > 0$. The holomorphic sectional curvature of $\mathbb{P}\mathcal{H}$, as in Definition 3.1.31, is strictly constant and identically equal to $1/\lambda$.

The core proof of this proposition is found in [6, Section 4.5., p.120]. The rescaling fixed in Equation (3.23) was chosen to reproduce the Schrödinger flow, but it has an equally direct geometric consequence for the curvature computed in Proposition 3.3.8.

Corollary 3.3.9 (The curvature of $\mathbb{P}\mathcal{H}$ fixes $\hbar$)**.** *The physical Kähler structure of Equations* (3.24) *has constant holomorphic sectional curvature* $1/\hbar$*.*

Proof. The local Kähler potential $K = \hbar \ln\left(1 + \|z\|^2\right)$, obtained by rescaling as in Equation (3.24), is the potential K_λ of Proposition 3.3.8 for $\lambda = \hbar$, whose holomorphic sectional curvature is $1/\lambda = 1/\hbar$. □

This result can be generalized to any projective space. The central concept presented in the following theorem is the uniqueness, up to isomorphisms, of the projective Hilbert space formalism. The key importance of this next theorem will become clear during the constructions in the next sections.

Theorem 3.5 (Hawley–Igusa)**.** *Up to isomorphism, projective spaces are the only connected, simply connected, complete Kähler manifolds of constant, positive holomorphic sectional curvature.*

The proof of this theorem is presented in [**KobayashiNomizu1969**].

3.4 Riemannian Metric and Uncertainty Relations

While the symplectic form ω dictates the Hamiltonian dynamics of the quantum system, as discussed in Section 3.3, the Riemannian metric g encodes its statistical structure. The metric allows for the calculation of dispersion, distance, and transition probability on the state manifold.

Following Definition 3.1.13 applied to the Kähler manifold $(\mathbb{P}\mathcal{H}, \omega, g)$, the state space carries a symmetric Riemann bracket. For any two smooth functions $f, k \in C^\infty(\mathbb{P}\mathcal{H}, \mathbb{R})$, this bracket is defined by $((f, k)) = \langle df, Gdk\rangle$, where G is the musical isomorphism associated with the Fubini–Study metric g. We use the notation $((\cdot,\cdot))$, whether the arguments are general smooth functions on $\mathbb{P}\mathcal{H}$ or mean value functions of quantum observables.

In the next lemma, we show that exploiting the linear structure of the ambient Hilbert space $\mathcal{H}$ helps us to compute the intrinsic Kähler structures. We demonstrate how the Riemannian metric and the symplectic form on the tangent space of $\mathbb{P}\mathcal{H}$ can be recovered from the standard Hilbert inner product by simply projecting out the unphysical scale and phase degrees of freedom.

Lemma 3.4.1 (Metric and form via ambient representatives). *Let $\mathbb{P}\mathcal{H}$ be a projective Hilbert space as in Definition 3.2.2. Let $\psi = [\Psi] \in \mathbb{P}\mathcal{H}$, $\|\Psi\| = 1$, and let $\eta, \eta' \in \mathcal{H}$ be arbitrary. Define $\eta_\Psi := \eta - (\Psi, \eta)\Psi \in \Psi^\perp$ for the component of η tangent to $\mathbb{P}\mathcal{H}$ at ψ, under the identification $T_\psi\mathbb{P}\mathcal{H} \cong \Psi^\perp$ of Proposition 3.2.6. Then*

$$(\eta_\Psi, \eta'_\Psi) = (\eta, \eta') - (\eta, \Psi)(\Psi, \eta'). \tag{3.25}$$

and, by Equation (3.24)*,*

$$g(\eta_\Psi, \eta'_\Psi) = 2\hbar\,\mathrm{Re}\big[(\eta, \eta') - (\eta, \Psi)(\Psi, \eta')\big], \qquad \omega(\eta_\Psi, \eta'_\Psi) = 2\hbar\,\mathrm{Im}\big[(\eta, \eta') - (\eta, \Psi)(\Psi, \eta')\big]. \tag{3.26}$$

Proof. Since $\|\Psi\| = 1$, η_Ψ is the orthogonal projection of η onto $\Psi^\perp$, and likewise for η'_Ψ. Denoting $a := (\Psi, \eta)$, $b := (\Psi, \eta')$ and expanding by conjugate-linearity in the first argument yields

$$(\eta_\Psi, \eta'_\Psi) = (\eta - a\Psi, \eta' - b\Psi) = (\eta, \eta') - b(\eta, \Psi) - \bar{a}\,b + \bar{a}\,b = (\eta, \eta') - (\eta, \Psi)(\Psi, \eta'),$$

which is Equation (3.25). Equation (3.26) follows applying Equation (3.24). □

Remark 3.4.2. In particular, for $A \in \mathcal{B}(\mathcal{H})$ selfadjoint, $x = \Psi \in \mathbb{S}\mathcal{H}$ normalized, and $\eta := -\frac{i}{\hbar}Ax$, one has $(\Psi, \eta) = -\frac{i}{\hbar}\langle A\rangle([x])$ and $\eta_\Psi = -\frac{i}{\hbar}(A - \langle A\rangle)x$. This recovers the vector field $X_{\langle A\rangle}([x])$ of Section 3.3.

The following proposition establishes how the Jordan product, as in Definition **??**, of two selfadjoint operators is connected with the Riemannian structure of $\mathbb{P}\mathcal{H}$, requiring no reference to the underlying linear Hilbert space.

Proposition 3.4.3 (Intrinsic Jordan decomposition). *Let $\mathbb{P}\mathcal{H}$ be a projective Hilbert space as in Definition 3.2.2. Let $A, K \in \mathcal{B}(\mathcal{H})$ be selfadjoint operators, as in Definition* **??***, and let $f = \langle A\rangle$ and $k = \langle K\rangle$ be their corresponding mean value functions on $\mathbb{P}\mathcal{H}$, as in Definition 3.3.1. The mean value of their Jordan product $A \circ K$ is uniquely determined by the Riemann bracket on $\mathbb{P}\mathcal{H}$ via the relation*

$$\langle A \circ K\rangle = \frac{\hbar}{2}((f, k)) + fk. \tag{3.27}$$

Proof. Fix $[x] \in \mathbb{P}\mathcal{H}$ and a normalized representative $\Psi = x \in \mathbb{S}\mathcal{H}$. By the remark following Lemma 3.4.1, X_f and X_k are represented, under $T_{[x]}\mathbb{P}\mathcal{H} \cong x^\perp$, by the projections of $\eta := -\frac{i}{\hbar}Ax$ and $\eta' := -\frac{i}{\hbar}Kx$. Kähler compatibility, as in Remark 3.1.15, yields $((f, k)) = g(X_f, X_k) = g(\eta_x, \eta'_x)$, then Lemma 3.4.1 applies. Using $(\eta, x) = \frac{i}{\hbar}\langle A\rangle([x])$, $(x, \eta') = -\frac{i}{\hbar}\langle K\rangle([x])$, selfadjointness of A, K, and

$$\mathrm{Re}(Ax, Kx) = \big(x, \tfrac{1}{2}(AK + KA)x\big) = \langle A \circ K\rangle([x]),$$

we have that

$$((f, k))_\omega([x]) = 2\hbar\,\mathrm{Re}\left[\frac{1}{\hbar^2}(Ax, Kx) - \frac{1}{\hbar^2}\langle A\rangle([x])\langle K\rangle([x])\right] = \frac{2}{\hbar}\big[\langle A \circ K\rangle([x]) - f([x])k([x])\big],$$

which is Equation (3.27). Equation (3.28) is the case $K = A$, since $A \circ A = A^2$. □

The dispersions of two observable functions are controlled jointly, through the metric and the symplectic form, by the Kähler counterpart of the Robertson–Schrödinger uncertainty relation.

Definition 3.4.4 (Dispersion). *Let $f \in \mathcal{K}(\mathbb{P}\mathcal{H},\mathbb{R})$ be a Kählerian function, as per Definition 3.1.19, and let $p \in \mathbb{P}\mathcal{H}$. The* dispersion *of f at p is*

$$\Delta f(p) := \sqrt{\hbar/2}\,\sqrt{((f,f))(p)}. \tag{3.28}$$

If $f = \langle A\rangle$ for a selfadjoint $A \in \mathcal{B}(\mathcal{H})$, $\Delta f(p)$ coincides with the usual quantum-mechanical standard deviation of A in the state p, by Remark 3.4.5 below.

Remark 3.4.5. Setting $A = K$ and $f = k = \langle A\rangle$ in Proposition 3.4.3 yields $\langle A^2\rangle = \frac{\hbar}{2}((f,f)) + f^2$, that is,

$$((f,f)) = \frac{2}{\hbar}\Big(\langle A^2\rangle - \langle A\rangle^2\Big) = \frac{2}{\hbar}(\Delta f)^2, \tag{3.29}$$

consistently with Definition 3.4.4. While $((f,k))$ alone does not generally represent the expectation value of a single observable, it serves as the geometric formulation of the quantum variance between f and k at any given state $p \in \mathbb{P}\mathcal{H}$.

This geometric interpretation of variances finds the way to formulate the most general uncertainty relations without invoking the linear structure of $\mathcal{H}$.

Theorem 3.6 (Robertson–Schrödinger relation, geometric form). *Let $\mathbb{P}\mathcal{H}$ be a projective Hilbert space as in Definition 3.2.2. For any pair of observable functions $f, k \in \mathcal{K}(\mathbb{P}\mathcal{H},\mathbb{R})$, as in Definition 3.5.1, their statistical dispersions evaluated at any state $p \in \mathbb{P}\mathcal{H}$ satisfy the bound:*

$$(\Delta f)(\Delta k) \geq \frac{\hbar}{2}\sqrt{\{f,k\}^2 + ((f,k))^2}. \tag{3.30}$$

Proof. Fix $[x] \in \mathbb{P}\mathcal{H}$. Since $X_f, X_k \in x^{\perp}$ are already tangent, Equation (3.24) yields

$$((f,k)) + i\,\{f,k\} = g(X_f,X_k) + i\,\omega(X_f,X_k) = 2\hbar\,(X_f,X_k).$$

By Equation (3.28), $\|X_f\|^2 = (X_f,X_f) = \frac{1}{2\hbar}((f,f)) = \frac{(\Delta f)^2}{\hbar^2}$, and likewise for k. The Cauchy–Schwarz inequality on $\mathcal{H}$ then yields

$$\sqrt{((f,k))^2 + \{f,k\}^2} = 2\hbar\,|(X_f,X_k)| \leq 2\hbar\,\|X_f\|\,\|X_k\| = \frac{2}{\hbar}(\Delta f)(\Delta k),$$

which is the stated inequality. □

Remark 3.4.6. The weaker Heisenberg relation $(\Delta f)(\Delta k) \geq \frac{\hbar}{2}|\{f,k\}|$ follows dropping the Riemann bracket term. Equality in Theorem 3.6 holds at $[x] \in \mathbb{P}\mathcal{H}$ if and only if X_f and X_k are proportional over $\mathbb{C}$ there, $X_f = aX_k + b\,JX_k$ for some $a, b \in \mathbb{R}$. The extreme cases $b = 0$ and $a = 0$ correspond, respectively, to the vanishing of the Poisson bracket and of the covariance $((f,k))$ at $[x]$.

Example 3.4.7. For $f = \langle\sigma_x\rangle$, $k = \langle\sigma_y\rangle$ on the Bloch sphere, at the state $[x] = [(1,0)^T]$ one computes $\{f,k\} = \frac{2}{\hbar}\langle\sigma_z\rangle = \frac{2}{\hbar}$ and $((f,k)) = 0$, so Theorem 3.6 yields $(\Delta f)(\Delta k) \geq 1$, saturated at this state since $\langle\sigma_x\rangle = \langle\sigma_y\rangle = 0$ there force $\Delta f = \Delta k = 1$, matching the standard spin uncertainty relation.

A related special case, isolating the variance of a single Kählerian function against its own Hamiltonian flow, is worth recording explicitly.

Example 3.4.8 (Anandan–Aharonov relation). For a bounded Hamiltonian $H \in \mathcal{B}(\mathcal{H})$ with observable function $h = \langle H\rangle \in \mathcal{K}(\mathbb{P}\mathcal{H},\mathbb{R})$, the energy variance takes the form $(\Delta h)^2 = \frac{\hbar}{2}((h,h))$. By the Kähler compatibility outlined in Remark 3.1.15, $((h,h)) = g(X_h,X_h)$. Thus, the energy uncertainty Δh equals, up to a constant, the Riemannian length of the Hamiltonian vector field generating time evolution. The energy dispersion dictates the Riemannian speed at which the state traverses the projective manifold $\mathbb{P}\mathcal{H}$ during unitary dynamics.

Geodesics and Riemannian Distance

The Riemannian metric g studied so far entails the infinitesimal statistical structure of $\mathbb{P}\mathcal{H}$. That encodes the local dispersion and variance of observables within the tangent space. Instead, capturing the macroscopic distinguishability between quantum states, we must integrate this structure across the manifold. The global physical role of the metric is thus expressed through the distance it induces. The extremal curves that minimize this geometric length are called geodesics, which we define as follows.

Definition 3.4.9 (Geodesic). *Let $(\mathcal{M}, J, g)$ be a Kähler manifold, as per Definition 3.1.10. For a piecewise C^1 curve $\gamma : [a, b] \to \mathcal{M}$, its length is*

$$L(\gamma) := \int_a^b \sqrt{g(\dot{\gamma}(t), \dot{\gamma}(t))}\, dt. \tag{3.31}$$

A curve $\gamma : [a, b] \to \mathcal{M}$, parametrized proportionally to arc length, is a geodesic *if every $t_0 \in [a, b]$ has a neighborhood $[t_0 - \varepsilon, t_0 + \varepsilon] \cap [a, b]$ on which γ minimizes L among piecewise C^1 curves with the same endpoints.*

Geodesics are the Riemannian analogue of straight lines, and their lengths define a genuine distance function on $\mathcal{M}$, compatible with its manifold topology.

Definition 3.4.10 (Riemannian distance). *Let $(\mathcal{M}, J, g)$ be a Kähler manifold, as per Definition 3.1.10. The Riemannian distance between $p, q \in \mathcal{M}$ is*

$$d_g(p, q) := \inf_{\gamma} L(\gamma), \tag{3.32}$$

where the infimum runs over piecewise C^1 curves $\gamma : [a, b] \to \mathcal{M}$ with $\gamma(a) = p$, $\gamma(b) = q$, and L is as in Definition 3.4.9.

Remark 3.4.11. Note that, in Definition 3.4.10, if the manifold $\mathcal{M}$ is connected, d_g is a metric inducing the manifold topology as per Proposition 3.2.7. Moreover, when attained, the infimum is realized by a geodesic of Definition 3.4.9.

Among submanifolds of $\mathcal{M}$, those on which the ambient and intrinsic geodesics coincide play a distinguished role. For these, the intrinsic distance agrees with the distance measured in the ambient manifold.

Definition 3.4.12 (Totally geodesic submanifold). *Let $(\mathcal{M}, J, g)$ be a Kähler manifold, as per Definition 3.1.10, and let $N \subset \mathcal{M}$ be a submanifold, equipped with the induced metric $g|_N$. N is* totally geodesic *if every geodesic of $(N, g|_N)$, as per Definition 3.4.9, is also a geodesic of $(\mathcal{M}, g)$.*

Remark 3.4.13. If $N \subset \mathcal{M}$ is totally geodesic as per Definition 3.4.12 and $\mathcal{M}$ is connected, the Riemannian distance $d_{g|_N}$ induced on N by Definition 3.4.10 coincides with the restriction of d_g to $N \times N$.

Transition Probabilities

Beyond its algebraic role in defining the quantum variance of specific observable functions, the Riemannian metric g has an operational meaning independent of any fixed physical quantity. In fact, it quantifies the statistical distinguishability of infinitesimally close pure states. When evaluating a continuous parameterization of the state space, the metric tensor yields the quantum Fisher information. That sets the geometric limit on how nearby states can be resolved by any experimental measurement.

Theorem 3.7 (Provost–Vallee). *Let $\mathcal{H}$ be a Hilbert space as in Definition* **??** *and $\mathbb{P}\mathcal{H}$ be its projective space as per Definition 3.2.2. Let $\theta \in \mathbb{R} \mapsto x(\theta) \in \mathcal{H} \setminus \{0\}$ be a smooth curve, and $\psi(\theta) := [x(\theta)] \in \mathbb{P}\mathcal{H}$. Setting the velocity $\dot{x} := \partial_\theta x$ yields*

$$g\left(\frac{d\psi}{d\theta}, \frac{d\psi}{d\theta}\right) = 2\hbar\left(\frac{(\dot{x}, \dot{x})}{(x, x)} - \frac{(x, \dot{x})(\dot{x}, x)}{(x, x)^2}\right). \tag{3.33}$$

This result is not proved here. The reader is referred to the original article by Provost and Vallee in [7, Section 2].

Remark 3.4.14. The scalar quantity

$$\mathcal{F}(\theta) := 4\left(\frac{(\dot{x},\dot{x})}{(x,x)} - \frac{(x,\dot{x})(\dot{x},x)}{(x,x)^2}\right) \tag{3.34}$$

is the one-dimensional quantum Fisher information of the family of pure states $\psi(\theta)$ with respect to the parameter θ. Theorem 3.7 identifies it with the pulled-back metric, $g(\dot{\psi},\dot{\psi}) = \frac{\hbar}{2}\mathcal{F}(\theta)$. The Riemannian line element of $\mathbb{P}\mathcal{H}$ is, up to the constant $\hbar/2$, the statistical distance between infinitesimally separated pure states, measuring the maximum achievable precision in estimating the physical parameter θ.

We clarify its meaning through the next example.

Example 3.4.15 (Fisher Information of a Rotation). Let $\mathcal{H} = \mathbb{C}^2$ and consider a single-parameter family of pure states defined by the normalized vector $x(\theta) = (\cos(\theta/2), \sin(\theta/2))^T$. This corresponds to a one-parameter unitary rotation of the state vector. The parameter derivative is

$$\dot{x}(\theta) = \frac{1}{2}\begin{pmatrix} -\sin(\theta/2) \\ \cos(\theta/2) \end{pmatrix}. \tag{3.35}$$

The inner products evaluate to $(\dot{x},\dot{x}) = \frac{1}{4}(\sin^2(\theta/2)+\cos^2(\theta/2)) = \frac{1}{4}$ and $(x,\dot{x}) = -\frac{1}{2}\cos(\theta/2)\sin(\theta/2)+\frac{1}{2}\sin(\theta/2)\cos(\theta/2) = 0$. Substituting these into the quantum Fisher information formula yields

$$\mathcal{F}(\theta) = 4\left(\frac{1}{4} - 0\right) = 1. \tag{3.36}$$

The pulled-back Fubini–Study metric evaluates to $g\left(\frac{d\psi}{d\theta}, \frac{d\psi}{d\theta}\right) = \frac{\hbar}{2}$. The quantum Fisher information being strictly constant and equal to 1 demonstrates that the parameter θ acts directly as the geodesic arc length on the Bloch sphere up to a scaling factor, meaning the statistical distinguishability of the state accumulates uniformly with respect to θ.

The Geometrical Born Rule

The metric g extends beyond infinitesimal statistical dispersion to govern the macroscopic distinguishability of quantum states. Integrating the metric distance across the manifold yields the postulate of quantum measurement.

Theorem 3.8 (Geometrical Born Rule). *Let $d_g : \mathbb{P}\mathcal{H} \times \mathbb{P}\mathcal{H} \to \mathbb{R}$ denote the Riemannian distance, as per Definition 3.4.10, induced by the Fubini–Study metric g on $\mathbb{P}\mathcal{H}$. For any two pure states $[x],[y] \in \mathbb{P}\mathcal{H}$, the quantum mechanical transition probability $P([x],[y])$ satisfies*

$$P([x],[y]) := \frac{|(x,y)|^2}{\|x\|^2\|y\|^2} = \cos^2\left(\frac{d_g([x],[y])}{\sqrt{2\hbar}}\right). \tag{3.37}$$

We will not present the proof of this statement. For further readings, we refer to [8, Theorem II.2].

Example 3.4.16 (Qubit). For $\mathcal{H} = \mathbb{C}^2$, the projective space $\mathbb{P}\mathcal{H}$ is identified with $\mathbb{C}P^1$, the Bloch sphere of radius $\sqrt{\hbar/2}$ in the Fubini–Study metric g of (3.24). Antipodal points on the sphere correspond to orthogonal states $[x],[y]$, for which $P([x],[y]) = |(x,y)|^2/(\|x\|^2\|y\|^2) = 0$. By Theorem 3.8 this forces $d_g([x],[y]) = \pi\sqrt{\hbar/2}$, the diameter of the sphere, matching the maximal geodesic distance recorded in the remark following Theorem 3.5. Conversely, identical states sit at $d_g = 0$, where $P = 1$, then $\cos^2$ interpolates continuously between certainty and mutual exclusivity as the geodesic distance ranges over $[0, \pi\sqrt{\hbar/2}]$.

Remark 3.4.17. The metric g completely determines the probabilistic structure of quantum mechanics. Orthogonal states, representing mutually exclusive measurement outcomes, correspond to points separated by the maximum geodesic distance $d_g = \pi\sqrt{\hbar/2}$.

3.5 Observable Functions on Kähler manifolds

In classical mechanics, the space of physical observables is identified with the entire commutative algebra of smooth real-valued functions on the symplectic phase space. In the quantum setting, by contrast, the constraints set by the symplectic structure ω and the Riemannian metric g define the admissible observables. The physical observables are those functions whose Hamiltonian flow respects the full Kähler geometry of the state manifold.

Definition 3.5.1 (Observable function). *A smooth function $f : \mathbb{P}\mathcal{H} \to \mathbb{R}$ is an* observable function *if there exists a selfadjoint operator $A \in \mathcal{B}(\mathcal{H})$ such that $f = \langle A \rangle$.*

An equivalent definition can be stated using a dynamical consideration based on Section 3.3. By Proposition 3.3.6, the flow generated by a selfadjoint operator A via the Schrödinger equation acts on $\mathbb{P}\mathcal{H}$ as a continuous one-parameter group of Kähler isomorphisms. Consequently, the Hamiltonian vector field X_f of an observable function $f = \langle A \rangle$ preserves the symplectic form ω and also the Fubini–Study metric g.

Definition 3.5.2 (Killing vector field). *Let $(\mathcal{M}, J, g)$ be a Kähler manifold, as per Definition 3.1.10. A vector field $X \in \mathfrak{X}(\mathcal{M})$ is a* Killing vector field *if the Lie derivative of the metric tensor g with respect to X vanishes identically on $\mathcal{M}$:*

$$\mathcal{L}_X g = 0. \tag{3.38}$$

Hence, we can characterize the observables as the functions which generate a Killing Hamiltonian vector field.

Theorem 3.9 (Geometric characterization of observables). *Given a projective Hilbert space $\mathbb{P}\mathcal{H}$ as per Definition 3.2.2. A smooth function $f \in C^\infty(\mathbb{P}\mathcal{H}, \mathbb{R})$ is an observable function, as in Definition 3.5.1, if and only if its Hamiltonian vector field X_f is a Killing vector field for the Fubini–Study metric g, as per Definition 3.5.2.*

The proof of this statement is shown in [8, p.13]. For instance, we present an example of a Killing vector field on the Bloch sphere.

Example 3.5.3. On the Bloch sphere $\mathbb{P}\mathcal{H} \cong \mathbb{C}P^1$ with the Fubini–Study metric g, the one-parameter group $[x] \mapsto [e^{-it\sigma_z/2}x]$ rotates the sphere about its polar axis at constant angular speed. Since rotations preserve the round metric, the generating vector field $X_{\langle \sigma_z \rangle}$ is Killing. This confirms, via Theorem 3.9, that $\langle \sigma_z \rangle$ is an observable function.

Remark 3.5.4 (Symmetry data and finite-dimensionality). A property of Killing vector fields on a connected Kähler manifold is their unique global reconstruction from the initial data consisting of their pointwise value and their first derivative at a single prescribed base point. Through the symplectic correspondence

$$\mathrm{Ham}(f) = I\, \mathrm{d}f.$$

This uniqueness principle entails that any observable function on the projective Hilbert space $\mathbb{P}\mathcal{H}$ is completely determined by its value, its exterior derivative, and its Hessian tensor, all evaluated at an arbitrary reference state $[x] \in \mathbb{P}\mathcal{H}$. Consequently, in the case where $\mathcal{H}$ is finite-dimensional, the vector space of such observables is likewise finite-dimensional. This stands in contrast with the infinite-dimensional structure of the standard algebra of smooth functions $C^\infty(\mathbb{P}\mathcal{H}, \mathbb{R})$, which exhibits no such finite parametrization.

Theorem 3.9 establishes that observables are the Kählerian functions in Definition 3.1.19 of the projective Hilbert space. They act as the generators of the structural symmetries of the state manifold. This geometric constraint allows the non-commutative algebraic operations of $\mathcal{B}(\mathcal{H})$ to be translated into geometric operations on $\mathbb{P}\mathcal{H}$.

Theorem 3.10 (Homomorphism of Lie–Jordan and Kähler algebras). *Let $(\mathcal{O}, \circ, \{\cdot,\cdot\})$ be a real Lie–Jordan algebra, as per Definition* **??**. *Let $\mathfrak{A}$ be the C^*-algebra generated by $\mathcal{O}$, as per Theorem* **??***, and let $\mathbb{P}\mathcal{H}_\alpha$ be the projective Hilbert space corresponding to a fixed sector of the pure state space $\mathcal{P}(\mathfrak{A})$, as per Theorem 3.4.*

The mean value map $A \mapsto \langle A \rangle$, as per Definition 3.3.1, on $\mathbb{P}\mathcal{H}_\alpha$, is a Lie–Jordan homomorphism between $\mathcal{O}$ and the algebra of Kählerian functions $\mathcal{K}(\mathbb{P}\mathcal{H}_\alpha, \mathbb{R})$, as per Definition 3.1.19. The map is an algebraic-geometric isomorphism if and only if the GNS representation of $\mathfrak{A}$ on the sector α, as per Theorem **??***, is faithful, as per Definition* **??**.

Proof. By Theorem **??**, the abstract Lie–Jordan algebra $\mathcal{O}$ embeds as the selfadjoint part of the C^*-algebra $\mathfrak{A}$. By Theorem 3.4, the pure state space $\mathcal{P}(\mathfrak{A})$ decomposes into a disjoint union of projective Hilbert spaces $\mathbb{P}\mathcal{H}_\alpha$, each endowed with an intrinsic Kähler structure (J,g,ω), as per Definition 3.1.10.

For a fixed sector $\mathbb{P}\mathcal{H}_\alpha$, an element $A \in \mathcal{O}$ defines a smooth real-valued expectation function $\langle A\rangle : \mathbb{P}\mathcal{H}_\alpha \to \mathbb{R}$. The algebraic operations defined on $\mathcal{O}$ correspond to the geometric structures on $\mathbb{P}\mathcal{H}_\alpha$ via the following evaluations:

1. **Lie structure (Symplectic form):** The Lie bracket $\{A,B\}$ in $\mathcal{O}$ corresponds to the Poisson bracket generated by the symplectic form ω, as per Definition 3.1.13. Applying Proposition 3.3.4 yields the exact relation:
$$\{\langle A\rangle,\langle B\rangle\} = \langle\{A,B\}\rangle. \tag{3.39}$$
2. **Jordan structure (Riemannian metric):** The Jordan product $A\circ B$ in $\mathcal{O}$ governs the statistical dispersion structure. This maps to the scaled Riemann bracket generated by the Fubini–Study metric g, as per Definition 3.1.13, isolating the $\circ_\hbar$ product by Proposition 3.4.3 and Definition 3.1.16:
$$\langle A\circ B\rangle = \langle A\rangle \circ_\hbar \langle B\rangle = \frac{\hbar}{2}((\langle A\rangle,\langle B\rangle)) + \langle A\rangle\langle B\rangle. \tag{3.40}$$
3. **Associative product (Kähler form):** The total associative product $AB = A\circ B + \frac{i\hbar}{2}\{A,B\}$, given by Equation (**??**), within the complexified enveloping algebra $\mathfrak{A}$ reconstructs the geometric Kähler product $*_\hbar$, as per Definition 3.1.16. Applying the linearity of the expectation value map on the components yields:
$$\begin{aligned}\langle AB\rangle &= \langle A\circ B\rangle + \frac{i\hbar}{2}\langle\{A,B\}\rangle \\ &= \left(\frac{\hbar}{2}((\langle A\rangle,\langle B\rangle)) + \langle A\rangle\langle B\rangle\right) + \frac{i\hbar}{2}\{\langle A\rangle,\langle B\rangle\} \\ &= \langle A\rangle *_\hbar \langle B\rangle.\end{aligned} \tag{3.41}$$

This establishes the structural homomorphism from the abstract algebra $\mathcal{O}$ into the geometric algebra on $\mathbb{P}\mathcal{H}_\alpha$. The kernel of this map coincides with the kernel of the GNS representation π_α associated with the sector α, as per Theorem **??**. Consequently, the map constitutes an exact isomorphism if and only if the specified sector provides a faithful representation of the observable algebra, as per Definition **??**. □

Before turning to an example, we note the simultaneous presence of the constant $\hbar$ in the geometric $*_\hbar$ product and in the rescaling of Equation (3.24). As we have seen, Kähler compatibility forces $\omega = 2\hbar\,\omega_0$, giving $g = 2\hbar\, g_0$ and $K_\hbar = \hbar\ln\big(1+\|z\|^2\big)$. In addition, the holomorphic sectional curvature becomes $1/\hbar$. Likewise, the Lie–Jordan algebra isomorphism requires $\circ_\nu$ and $*_\nu$ to be evaluated at $\nu=\hbar$.

Corollary 3.5.5 (Imposition of $\nu = \hbar$)**.** *Let $\mathbb{P}\mathcal{H}$ be a projective Hilbert space as in Definition 3.2.2 and $A,B\in\mathcal{B}(\mathcal{H})$ be selfadjoint operators. If the products, defined in Definition 3.1.16, $((\langle A\rangle,\langle B\rangle))$ and $\{\langle A\rangle,\langle B\rangle\}$ are not identically zero on $\mathbb{P}\mathcal{H}$, then*
$$\langle A\rangle *_\nu \langle B\rangle = \langle AB\rangle \ \text{ on } \mathbb{P}\mathcal{H} \qquad\Longleftrightarrow\qquad \nu=\hbar. \tag{3.42}$$

Proof. Fix the functions $f=\langle A\rangle$ and $k=\langle B\rangle$. By Equation (**??**), $AB = A\circ B + \frac{i\hbar}{2}\{A,B\}$, linearity of the mean value map yields
$$\langle AB\rangle = \langle A\circ B\rangle + \tfrac{i\hbar}{2}\langle\{A,B\}\rangle.$$
By Proposition 3.4.3, $\langle A\circ B\rangle = fk + \frac{\hbar}{2}((f,k))$, and by Proposition 3.3.4, $\langle\{A,B\}\rangle = \{f,k\}$. Hence
$$\langle AB\rangle = fk + \tfrac{\hbar}{2}((f,k)) + \tfrac{i\hbar}{2}\{f,k\}. \tag{3.43}$$
By Definition 3.1.16, $f *_\nu k = fk + \frac{\nu}{2}\big[((f,k)) + i\{f,k\}\big]$. Subtracting from (3.43),
$$f *_\nu k - \langle AB\rangle = \tfrac{\nu-\hbar}{2}\big[((f,k)) + i\{f,k\}\big].$$
Since $((f,k))$ and $\{f,k\}$ are real-valued, the bracket $((f,k)) + i\{f,k\}$ vanishes at $[x]\in\mathbb{P}\mathcal{H}$ if and only if both terms do there. By hypothesis $f *_\nu k = \langle AB\rangle$ on $\mathbb{P}\mathcal{H}$ forces $\nu=\hbar$. Conversely, $\nu=\hbar$ yields equality by (3.43). □

Requiring the Kähler product $*_\nu$ to reproduce the associative operator product AB on mean value functions, equivalently, requiring compatibility between the algebraic structure of $\mathcal{B}(\mathcal{H})$ and the Kähler geometry of $\mathbb{P}\mathcal{H}$, forces $\nu = \hbar$. The value of $\hbar$ is thus fixed by the same normalization $\lambda = 2\hbar$ of Equation (3.23) that makes the Hamiltonian flow reproduce the Schrödinger equation.

Example 3.5.6. Let $A = \sigma_x$, $B = \sigma_y$ be Pauli matrices on $\mathcal{H} = \mathbb{C}^2$. Direct computation yields

$$A \circ B = \tfrac{1}{2}(\sigma_x\sigma_y + \sigma_y\sigma_x) = 0 \quad , \quad \{A,B\} = \tfrac{1}{i\hbar}[\sigma_x,\sigma_y] = \tfrac{2}{\hbar}\sigma_z.$$

This entails that $AB = A \circ B + \frac{i\hbar}{2}\{A,B\} = i\sigma_z$, matching the familiar identity $\sigma_x\sigma_y = i\sigma_z$. Under the isomorphism of Theorem 3.10, this same identity is recovered geometrically as $\langle\sigma_x\rangle *_\hbar \langle\sigma_y\rangle = \langle i\sigma_z\rangle$ on the Bloch sphere $\mathbb{P}\mathcal{H} \cong \mathbb{C}P^1$, illustrating how the noncommutative operator product becomes the $*_\hbar$ product of Definition 3.1.16.

3.6 Quantum Symmetries and the Momentum Map

Continuous symmetries and conservation laws on $\mathbb{P}\mathcal{H}$ adopt classical Hamiltonian nomenclature. The momentum map, Definition **??**, can be translated smoothly, yet the domain imposes stricter constraints. In fact, quantum symmetries must preserve the full Kähler triad (ω, g, J), whereas classical ones only preserve ω. By Wigner's Theorem, [5, Theorem 12.11], continuous symmetry groups arise from strongly continuous unitary representations on $\mathcal{H}$. We construct the quantum momentum map, assigning an observable to each Lie-algebra generator. Classical divergences appear, non-exact ω requires simple connectivity of $\mathbb{P}\mathcal{H}$ for global map existence, while g-invariance forces Kählerian components. The Quantum Noether Theorem then proves that geometric conservation of these components along the flow is equivalent to the vanishing commutator $[\hat{f}, \hat{H}]$.

Proposition 3.6.1 (Quantum momentum map). *Let $G \subseteq U(\mathcal{H})$ be a connected Lie group, as in Definition* **??***, acting on $\mathbb{P}\mathcal{H}$, a projective Hilbert space, via projective unitary transformations, as in Definition* **??***. Let $\mathfrak{g}$ be its Lie algebra, as in Definition* **??***, consisting of skew-selfadjoint operators on $\mathcal{H}$. There exists a unique momentum map $\mu : \mathbb{P}\mathcal{H} \to \mathfrak{g}^*$, satisfying $d\langle\mu,\xi\rangle = \iota_{\xi_{\mathbb{P}\mathcal{H}}}\omega$ as per Definition* **??***. That is given pointwise by*

$$\langle\mu([x]),\xi\rangle = \langle i\hbar\xi\rangle([x]) = \frac{(x, i\hbar\xi x)}{\|x\|^2}, \tag{3.44}$$

where $i\hbar\xi$ is the corresponding selfadjoint generator in $\mathcal{B}(\mathcal{H})$, and $\langle\cdot\rangle$ is the mean value function, as in Definition 3.3.1.

Proof. Let $G \subseteq U(\mathcal{H})$ be a connected Lie group acting on $\mathbb{P}\mathcal{H}$ via projective unitary transformations. The Lie algebra $\mathfrak{g}$ consists of skew-adjoint operators. Hence, $i\hbar\xi$ is selfadjoint for every $\xi \in \mathfrak{g}$. By Remark **??**, the defining condition of the momentum map $d\langle\mu,\xi\rangle = \iota_{\xi_{\mathbb{P}\mathcal{H}}}\omega$ holds if and only if $X_{\langle\mu,\xi\rangle} = \xi_{\mathbb{P}\mathcal{H}}$. By Definition **??**, the infinitesimal generator $\xi_{\mathbb{P}\mathcal{H}}$ induces the one-parameter flow $\Phi_t([x]) = [\exp(t\xi)x]$ on $\mathbb{P}\mathcal{H}$. At the same time, by Proposition 3.3.2, the Hamiltonian vector field associated with the mean value function $\langle H\rangle$ of a selfadjoint operator H generates the flow $\Phi_t([x]) = [\exp(-itH/\hbar)x]$. Substituting $H = i\hbar\xi$, the Hamiltonian flow evaluates to $[\exp(t\xi)x]$. This establishes the vector field identity

$$X_{\langle i\hbar\xi\rangle} = \xi_{\mathbb{P}\mathcal{H}}.$$

Therefore, $\langle\mu([x]),\xi\rangle = \langle i\hbar\xi\rangle([x])$ satisfies the definition of the momentum map. □

The triviality of the $U(1)$ phase action on $\mathbb{P}\mathcal{H}$ offers an immediate consistency check for the formula just derived.

Example 3.6.2. Let $G = U(1)$ act on $\mathcal{H}$ by $\Psi \mapsto e^{i\theta}\Psi$, with Lie algebra generator $\xi = i \cdot \mathbb{1}$. Since this action is trivial on $\mathbb{P}\mathcal{H}$, because it fixes every ray $[\Psi]$, the induced vector field $\xi_{\mathbb{P}\mathcal{H}}$ vanishes identically. The momentum map of Proposition 3.6.1 reduces to the constant $\langle\mu([x]),\xi\rangle = \langle i\hbar\xi\rangle([x]) = -\hbar$, consistent with the overall phase of Ψ carrying no physical content on the projective state space.

This construction differs from its classical counterpart in two respects, both rooted in the distinctive topology and geometry of $\mathbb{P}\mathcal{H}$.

Remark 3.6.3 (Topological exactness and structural rigidity). The transition from classical phase spaces to $\mathbb{P}\mathcal{H}$ introduces two geometric distinctions regarding the momentum map.

1. *Non-exactness of the symplectic form.* Classical momentum maps are frequently constructed via the exactness of the symplectic form $\omega = -d\theta$, typical of cotangent spaces T^*Q. Conversely, the Fubini–Study form ω on $\mathbb{P}\mathcal{H}$ is closed but not exact. That is because for $\dim \mathcal{H} < \infty$, $\mathbb{P}\mathcal{H}$ is compact, strictly prohibiting exact symplectic forms. The existence of the global momentum map μ on $\mathbb{P}\mathcal{H}$ relies instead on the topological fact that $\mathbb{P}\mathcal{H}$ is simply connected.

2. *Metric constraints.* In classical mechanics, the component functions $\langle \mu, \xi \rangle$ are smooth real functions generating symplectic vector fields. In this framework, because the flow of $\xi_{\mathbb{P}\mathcal{H}}$ preserves the Riemannian metric g, Theorem 3.9 guarantees that the components of the momentum map are Kählerian functions, or equivalently Killing vector fields. The classical observable corresponding to a continuous symmetry direction ξ is identified with the expectation value of the quantum observable $i\hbar\xi$.

The classical Geometric Noether Theorem, Theorem **??**, translates directly into the quantum formalism. The structure of $\mathbb{P}\mathcal{H}$ uniquely identifies the vanishing of the classical Poisson bracket with the vanishing of the quantum commutator.

Theorem 3.11 (Quantum Noether Theorem)**.** *Let $\mathbb{P}\mathcal{H}$ be a projective Hilbert space and $H \in \mathcal{B}(\mathcal{H})$ be a selfadjoint Hamiltonian operator governing the dynamics as in Section 3.3. Let $G \subseteq U(\mathcal{H})$ be a connected Lie group, as in Definition* **??***, acting on $\mathbb{P}\mathcal{H}$ with momentum map $\mu : \mathbb{P}\mathcal{H} \to \mathfrak{g}^*$, as in Proposition 3.6.1. The momentum map is an integral of motion, meaning $\mu(\Phi_t([x])) = \mu([x])$ for all $t \in \mathbb{R}$, if and only if the algebra of generators strictly commutes with the Hamiltonian*

$$[H, i\hbar\xi] = 0 \qquad \forall \xi \in \mathfrak{g}. \tag{3.45}$$

Proof. By Theorem **??**, μ is conserved along the flow of $X_{\langle H \rangle}$ if and only if the Poisson bracket vanishes. That is $\{\langle H \rangle, \langle \mu, \xi \rangle\} = 0$ for all $\xi \in \mathfrak{g}$. Substituting the explicit form of the quantum momentum map from Proposition 3.6.1, the condition becomes $\{\langle H \rangle, \langle i\hbar\xi \rangle\} = 0$. Applying the algebraic-geometric correspondence of Proposition **??**, this symplectic bracket maps to the mean value of the operator commutator

$$\{\langle H \rangle, \langle i\hbar\xi \rangle\} = \left\langle \frac{1}{i\hbar}[H, i\hbar\xi] \right\rangle. \tag{3.46}$$

The function $\langle \frac{1}{i\hbar}[H, i\hbar\xi] \rangle$ vanishes identically on $\mathbb{P}\mathcal{H}$ if and only if the corresponding operator $[H, i\hbar\xi]$ is the zero operator in $\mathcal{B}(\mathcal{H})$. □

3.7 Algebraic-Geometric Axiomatization of Quantum Mechanics

Chapter **??** reconstructed quantum mechanics from operational data. It started with a C^*-algebra $\mathfrak{A}$ and a separating state space S, per Definition **??**, whose GNS representation, Theorem **??**, supplies a Hilbert space. However, neither a dynamics nor a reason for the Hamiltonian structure of the observable algebra is derived there. In fact, nothing in $\mathfrak{A}$ singles out a Poisson bracket among its automorphisms. This chapter closes both gaps geometrically. Theorem 3.4 shows that the pure states of $\mathfrak{A}$ already organize themselves into a disjoint union of projective Hilbert spaces $\mathbb{P}\mathcal{H}_\alpha$. Hence, fixing one sector lets the Kähler manifold $\mathbb{P}\mathcal{H}_\alpha$ of Section 3.2, rather than $\mathfrak{A}$ itself, generate the symplectic form that drives the dynamics and the Riemann bracket that reproduces the Jordan product. We observe that the mean value map $A \mapsto \langle A \rangle$ is an algebra isomorphism by Theorem 3.10. It entails that every geometric fact proved about $\mathbb{P}\mathcal{H}_\alpha$ is simultaneously a fact about $\mathfrak{A}$. We also note that the choice of $\mathbb{P}\mathcal{H}_\alpha$, rather than some other Kähler manifold, is forced by Hawley–Igusa, Theorem 3.5.
In this section, we construct an axiomatization of quantum theory based on the algebraic-geometric framework presented in this chapter and the previous one.

Pure state decomposition and the choice of a sector

By Theorem 3.4, the pure state space of $\mathfrak{A}$ decomposes as

$$\mathcal{P}(\mathfrak{A}) = \bigcup_{\alpha} \mathbb{P}\mathcal{H}_\alpha,$$

indexed by the unitary equivalence classes of irreducible representations of $\mathfrak{A}$, that is, by the, so called, superselection sectors. Every construction from Section 3.2 to Section 3.6 is carried out on a single sector. In fact, fixing α fixes an irreducible GNS Hilbert space $\mathcal{H}_\alpha$ and its projective space $\mathbb{P}\mathcal{H}_\alpha$. It is $\mathbb{P}\mathcal{H}_\alpha$, not $\mathcal{P}(\mathfrak{A})$, that carries the triad (J, g, ω) of Equation (3.24). For instance, if $\mathfrak{A} = \mathfrak{B}_0(\mathcal{H})$, then Proposition 3.2.3 forces the decomposition to the single term $\mathbb{P}\mathcal{H}_\alpha = \mathbb{P}\mathcal{H}$. For an algebra with different sectors, the construction below applies one by one. The sectors are merged only at the level of $\mathcal{P}(\mathfrak{A})$, not at the level of any single symplectic or Riemannian structure.

Remark 3.7.1 (Vector states and the geometric point of a pure state). For a pure state $\omega \in \mathcal{P}(\mathfrak{A})$ lying in sector $\mathbb{P}\mathcal{H}_\alpha$, the GNS cyclic vector $\Psi_\omega \in \mathcal{H}_\alpha$ determines the point $[\Psi_\omega] \in \mathbb{P}\mathcal{H}_\alpha$. The expectation value can be rewritten geometrically as

$$\omega(A) = \frac{(\Psi_\omega, A\Psi_\omega)}{\|\Psi_\omega\|^2} = \langle A\rangle([\Psi_\omega]), \qquad \forall A \in \mathfrak{A}. \tag{3.47}$$

Hence, an algebraic state is the same object as a point of $\mathbb{P}\mathcal{H}_\alpha$, and evaluating the state functional is evaluating the mean value function of Definition 3.3.1.

Physical observables

The point at which the algebraic formalism is transferred to the geometric one is Theorem 3.9. On the algebraic side, an observable is a selfadjoint element $A \in \mathfrak{A}$. Definition 3.5.1 transports this to $\mathbb{P}\mathcal{H}_\alpha$ as any smooth $f = \langle A\rangle$. Theorem 3.9 then characterizes the image of this map without reference back to $\mathcal{H}_\alpha$. At this point, f arises from a selfadjoint operator if and only if its Hamiltonian vector field X_f is a Killing field for g, that is, if and only if f is Kählerian as per Definition 3.1.19. This is the moment the algebraic notion of observable becomes a differential-geometric one, identified by a metric condition on $\mathbb{P}\mathcal{H}_\alpha$.

Compatibility via the Lie–Jordan–Kähler isomorphism

Once observables are identified with Kählerian functions, Theorem 3.10, the mean value map $A \mapsto \langle A\rangle$ is the bridge between the two frameworks. Theorem 3.10 is the statement of compatibility. In fact, we show that $A \mapsto \langle A\rangle$ is an isomorphism between the Lie–Jordan algebra $(\mathcal{O}, \circ, \{\cdot,\cdot\})$ of bounded selfadjoint operators on $\mathcal{H}$ and the algebra of Kählerian functions $(\mathcal{K}(\mathbb{P}\mathcal{H}, \mathbb{R}), \circ_\hbar, \{\cdot,\cdot\})$. Hence, through this isomorphism, the geometric Poisson bracket of Definition 3.1.13 is, sector by sector, analogous to the algebraic commutator.Likewise the Riemann bracket is the Jordan anticommutator. Complexifying and passing to the norm of Definition **??**, Theorem **??** upgrades this to an isometric isomorphism of C^*-algebras between $\mathfrak{A}$ restricted to the sector and $\mathcal{K}_{\mathfrak{A}}(\mathbb{P}\mathcal{H}, \mathbb{C})$ with the $*_\hbar$ product. Every geometric statement proved in this chapter about $(\mathbb{P}\mathcal{H}, \omega, g)$ is, through Theorem 3.10 and Theorem **??**, already a statement about $\mathfrak{A}$.

Dynamics

The Kähler structure resolves the first gap identified above. Proposition 3.3.2 shows that the Schrödinger flow can be rewritten as the Hamiltonian flow, with respect to ω, of the observable function $\langle H\rangle$. Hence, Schrödinger dynamics is a case of Hamilton's equations on the projective manifold $\mathbb{P}\mathcal{H}$. Proposition 3.3.6 supplies the converse. A continuous one-parameter group of transformations of $\mathbb{P}\mathcal{H}$ preserving the full triad (ω, g, J) is necessarily of Schrödinger form, generated by a selfadjoint operator unique up to an additive constant. Preserving g together with ω is the geometric counterpart of unitary evolution. The algebraic notion of this structural preservation is caputred through automorphisms of the algebra. However, the GNS framework presented does not identify a single particular way to define the Schrödinger automorphism. At the symmetry level, Theorem 3.11 uses this requirement to identify the vanishing Poisson bracket $\{\langle H\rangle, \langle \mu, \xi\rangle\}$ with the commutator condition $[H, i\hbar\xi] = 0$, term by term, through Theorem 3.10.

Hawley–Igusa and the constant $\hbar$

Hawley–Igusa Theorem 3.5 is what makes it consistent to read $\hbar$ as a curvature. Among connected, simply connected, complete Kähler manifolds, constant positive holomorphic sectional curvature determines the manifold up to isomorphism. The curvature is consequently an intrinsic invariant of $\mathbb{P}\mathcal{H}_\alpha$, not just the normalization chosen for g. Corollary 3.3.9 evaluates this invariant, for the physical structure of Equation (3.24), to $1/\hbar$, positive for every $\hbar > 0$. Hawley-Igusa Theorem 3.5 guarantees that this is the value forced once $\mathbb{P}\mathcal{H}_\alpha$ is required to be homogeneously curved. Every other occurrence of $\hbar$ in the chapter is then fixed as a function of this single curvature as shown in Corollary 3.3.9. In fact, the metric normalization has been fixed at $\lambda = 2\hbar$ by Equation (3.23). Moreover, the Hamiltonian flow that reproduce the Stone evolution $e^{-itH/\hbar}$ agrees with this normalization factor.

Once that the curvature has been fixed, the value $\nu = \hbar$ identified by Corollary 3.5.5 is the only normalization for which the geometric product $*_\nu$ reproduces the associative operator product AB. Hence, Schrödinger's equation and the Lie–Jordan algebra structure of $\mathfrak{A}$ are tied to a single geometric invariant of the state manifold.

Measurement

The Riemannian metric g that fixes the observable algebra also carries the entire probabilistic content of quantum mechanics, at two complementary scales. Infinitesimally, Definition 3.4.4 reads the dispersion of an observable off the Riemann bracket, $\Delta f = \sqrt{\hbar/2}\sqrt{((f,f))}$. Moreover, Theorem 3.6 shows that the Robertson–Schrödinger uncertainty relation represents the Cauchy–Schwarz inequality for g. Hence, uncertainty is a local, metric phenomenon on $\mathbb{P}\mathcal{H}_\alpha$.

Globally, integrating g along curves produces the geodesics of Definition 3.4.9 and the Riemannian distance d_g of Definition 3.4.10, and Theorem 3.8 identifies the transition probability $P([x],[y])$ with the function of the Riemannian metric $\cos^2\left(d_g([x],[y])/\sqrt{2\hbar}\right)$. We showed that the same metric that bounds simultaneous dispersion at a point also measures the macroscopic distinguishability of two states through the length of the geodesic joining them. The Born rule is, in this sense, the large-scale expression of the same Riemannian metric already responsible for the uncertainty relations.

Axiomatics of the Geometric Framework

The original formulation combined geometric postulates with derived theorems. A geometric axiomatization of a quantum theory removes any *a priori* dependence on the linear structure of a Hilbert space or the algebraic properties of an ambient C^*-algebra. The framework rests on four irreducible, independent postulates.

(G1) **State Space Manifold:** The pure state space is a connected, simply connected, complete Kähler manifold $(\mathcal{M}, J, g)$ of constant holomorphic sectional curvature equal to $1/\hbar$.

(G2) **Observables:** The physical observables correspond bijectively to the algebra of Kählerian functions $\mathcal{K}(\mathcal{M},\mathbb{R})$.

(G3) **Dynamics:** Time evolution is governed by the Hamiltonian flow of a distinguished observable function $h \in \mathcal{K}(\mathcal{M},\mathbb{R})$.

(G4) **Measurement:** The transition probability between two pure states $p, q \in \mathcal{M}$ is determined by their Riemannian geodesic distance $d_g(p,q)$ via the geometric Born rule $P(p,q) = \cos^2\left(\frac{d_g(p,q)}{\sqrt{2\hbar}}\right)$.

All other structures established in Sections 3.2 through 3.6 are mathematical derivations proceeding from these four postulates.

- **Fubini–Study Geometry:** Postulate (G1) forces the manifold $\mathcal{M}$, up to isomorphism, to be a complex projective space $\mathbb{P}\mathcal{H}$ equipped with the scaled Fubini–Study triad (J, g, ω), by the Hawley–Igusa Theorem 3.5.

- **Algebraic Isomorphism:** Postulate (G2), combined with the intrinsic Poisson and Riemann brackets of the Kähler manifold, structurally generates a Lie–Jordan algebra. Theorem 3.10 derives the exact algebraic-geometric isomorphism with the space of bounded selfadjoint operators.
- **Schrödinger Evolution:** Postulate (G3) guarantees that the dynamics consist of global Kähler isomorphisms. By Proposition 3.3.2, this flow uniquely reconstructs the unitary Stone evolution on the underlying Hilbert space, eliminating the need to postulate the Schrödinger equation.
- **Uncertainty Relations:** The Robertson–Schrödinger bounds derived in Theorem 3.6 follow intrinsically from the metric and symplectic structures fixed by (G1) and (G2), completely independent of the probabilistic postulate (G4).

The unified axiomatic system

Definition 3.7.2 (Algebraic-geometric quantum system). *An* algebraic-geometric quantum system *is a triple* $(\mathfrak{A}, S, H)$ *defined by the following minimal axioms.*

(i) ***Algebraic Core:*** *$\mathfrak{A}$ is a unital C^*-algebra and S is a separating space of states, as per Definition* **??**.

(ii) ***Geometric Sectors:*** *The pure state space $\mathcal{P}(\mathfrak{A})$ decomposes into disjoint superselection sectors. Each sector α corresponds uniquely to a connected, simply connected, complete Kähler manifold $(\mathcal{M}_\alpha, J_\alpha, g_\alpha)$ of constant holomorphic sectional curvature $1/\hbar$.*

(iii) ***Observable Isomorphism:*** *For each sector α, the expectation value map $A \mapsto \langle A\rangle$ is a Lie–Jordan isomorphism between the selfadjoint elements $\mathfrak{A}_{sa}$ and the Kählerian functions $\mathcal{K}(\mathcal{M}_\alpha, \mathbb{R})$.*

(iv) ***Dynamical Equivalence:*** *Time evolution is governed by a selfadjoint Hamiltonian $H \in \mathfrak{A}_{sa}$. The corresponding observable function $h = \langle H\rangle$ generates a Hamiltonian flow on $\mathcal{M}_\alpha$ map to the algebraic automorphism $A \mapsto e^{itH/\hbar}Ae^{-itH/\hbar}$.*

(v) ***Measurement Postulate:*** *Transition probabilities on each $\mathcal{M}_\alpha$ obey the geometric Born rule, as per Theorem 3.8.*

Items (iv) and (v) of the Dirac–Von Neumann formulation are eliminated because they do not represent the modeling of a general quantum system but a specific one. The preservation of the Kähler structure is a differential-geometric consequence of the Hamiltonian flow specified in (iv). the structural matching of algebraic products is mathematically absorbed into the isomorphism requirement of (iii).

Remark 3.7.3 (Purification and stratified spaces). The space of density operators does not constitute a global smooth manifold. It forms a stratified space where the dimension of the local tangent space depends on the rank of the density matrix. Differential geometric structures, such as the symplectic form and the Riemannian metric, become singular across boundaries where the matrix rank changes.

The Gelfand–Naimark–Segal (GNS) construction circumvents this dimensional discontinuity via algebraic purification. By Theorem **??**, any state ω on the observable algebra $\mathcal{A}$, including mixed states, induces a Hilbert space $\mathcal{H}_\omega$ containing a cyclic vector Ψ_ω. Although Proposition **??** dictates that the representation π_ω is reducible when ω is mixed, the operational state is identically recovered as a pure vector state evaluation: $\omega(A) = (\Psi_\omega, \pi_\omega(A)\Psi_\omega)$. This maps the singular mixed state to the ray $[\Psi_\omega]$ in the corresponding projective space $\mathbb{P}\mathcal{H}_\omega$. Since $\mathbb{P}\mathcal{H}_\omega$ is a connected, complete Kähler manifold regardless of the reducibility of π_ω, the Kähler geometry is globally recovered, bypassing the rank-induced singularities of the density operator space. For a classification of these stratified manifolds, see [6].

3.8 Example III: Dynamics and Symmetries on the Bloch Sphere

This example returns to the quantum spin system of Section **??**, there reconstructed algebraically through the GNS representation of the $\mathfrak{su}(2)$-algebra generated by J_1, J_2, J_3. The same Pauli algebra and the same Bloch ball of Equation (**??**) reappear here as the projective Hilbert space $\mathbb{P}\mathcal{H}$ for $\mathcal{H} = \mathbb{C}^2$, now carrying the Kähler structure of Section 3.2. This finite-dimensional instance of the projective Hilbert space provides a concrete test case for the geometric apparatus developed in this chapter, from the Kähler structure of Section 3.2 to the momentum map of Section 3.6. Working through this example in detail also illustrates, on the same $\mathfrak{su}(2)$ Lie algebra already treated classically in Section **??**, how symmetries of a quantum system translate, through the Quantum Noether Theorem 3.11, into conserved quantities within the geometric language of $\mathbb{P}\mathcal{H}$.

Construction of the Kähler Manifold

First, we equip $\mathbb{P}\mathcal{H}(\mathbb{C}^2)$ with coordinates and explicitly construct its Kähler structure. Let $z = (z_0, z_1)^T \in \mathbb{C}^2 \setminus \{0\}$ denote a pure state vector, and let U_0 be the standard affine chart defined by the condition $z_0 \neq 0$. Introducing the local holomorphic coordinate $w := z_1/z_0 \in \mathbb{C}$, the state is represented on U_0 by the unnormalized vector $\Phi = (1, w)^T \in \mathbb{C}^2$, of squared norm $\|\Phi\|^2 = 1 + w\bar{w}$. By Equation (3.24), the local Kähler potential on U_0 is

$$K(w, \bar{w}) = \hbar \ln(1 + w\bar{w}). \tag{3.48}$$

Differentiating Equation (3.48) as per Theorem 3.3 yields the metric component

$$g_{w\bar{w}} = \partial_{\bar{w}} \partial_w K = \frac{\hbar}{(1 + |w|^2)^2}, \tag{3.49}$$

positive on all of U_0. It descends that the fundamental form

$$\boxed{\omega = i\, g_{w\bar{w}}\, dw \wedge d\bar{w} = \frac{i\hbar}{(1 + |w|^2)^2}\, dw \wedge d\bar{w}} \tag{3.50}$$

is nondegenerate. Since $\mathbb{P}\mathcal{H}(\mathbb{C}^2)$ has real dimension two, ω is a top-degree form on U_0. It descends that $d\omega = 0$ identically, so that, together with Equation (3.49), $(\mathbb{P}\mathcal{H}(\mathbb{C}^2), J, g)$ is a Kähler manifold as per Definition 3.1.10, in agreement with the general construction of Section 3.2.
The Bloch vector components $\mathbf{r} = (r_1, r_2, r_3) \in \mathbb{R}^3$, identified on the pure state boundary of the Bloch ball with the mean value functions $r_i := \langle \sigma_i \rangle$, constitute a set of global smooth functions on $\mathbb{P}\mathcal{H}(\mathbb{C}^2)$. On the local chart U_0 with coordinate $w = z_1/z_0$ and normalized representative $\Psi = (1 + |w|^2)^{-1/2}(1, w)^T$, they evaluate to the smooth functions

$$r_1 = \frac{w + \bar{w}}{1 + |w|^2}, \qquad r_2 = i\frac{\bar{w} - w}{1 + |w|^2}, \qquad r_3 = \frac{1 - |w|^2}{1 + |w|^2}. \tag{3.51}$$

The symplectic form $\omega = i g_{w\bar{w}} dw \wedge d\bar{w}$ defined in Equation (3.50) induces the Poisson bivector on U_0. For smooth functions $f, k \in C^\infty(U_0, \mathbb{R})$, the Poisson bracket is generated by the inverse of the symplectic matrix

$$\{f, k\} = \frac{1}{i g_{w\bar{w}}} \left(\frac{\partial f}{\partial w} \frac{\partial k}{\partial \bar{w}} - \frac{\partial f}{\partial \bar{w}} \frac{\partial k}{\partial w} \right) = \frac{(1 + |w|^2)^2}{i\hbar} \left(\frac{\partial f}{\partial w} \frac{\partial k}{\partial \bar{w}} - \frac{\partial f}{\partial \bar{w}} \frac{\partial k}{\partial w} \right). \tag{3.52}$$

Evaluating the partial derivatives of the mean value functions r_1 and r_2 yields

$$\frac{\partial r_1}{\partial w} = \frac{1 - \bar{w}^2}{(1 + |w|^2)^2}, \qquad \frac{\partial r_1}{\partial \bar{w}} = \frac{1 - w^2}{(1 + |w|^2)^2}, \tag{3.53}$$

$$\frac{\partial r_2}{\partial w} = -i\frac{1 + \bar{w}^2}{(1 + |w|^2)^2}, \qquad \frac{\partial r_2}{\partial \bar{w}} = i\frac{1 + w^2}{(1 + |w|^2)^2}. \tag{3.54}$$

Substitution into the Poisson bracket exhibits the algebraic cancellation.

$$
\begin{aligned}
\{r_1, r_2\} &= \frac{(1+|w|^2)^2}{i\hbar}\left[\left(\frac{1-\bar{w}^2}{(1+|w|^2)^2}\right)\left(i\frac{1+w^2}{(1+|w|^2)^2}\right)-\left(\frac{1-w^2}{(1+|w|^2)^2}\right)\left(-i\frac{1+\bar{w}^2}{(1+|w|^2)^2}\right)\right]\\
&= \frac{1}{i\hbar(1+|w|^2)^2}\left[i(1+w^2-\bar{w}^2-|w|^4)+i(1+\bar{w}^2-w^2-|w|^4)\right]\\
&= \frac{2(1-|w|^4)}{\hbar(1+|w|^2)^2} = \frac{2}{\hbar}\frac{1-|w|^2}{1+|w|^2}. \qquad (3.55)
\end{aligned}
$$

Comparing this scalar field with the structural definition of r_3 establishes the geometric derivation of the algebraic commutator, $\{r_1, r_2\} = \frac{2}{\hbar} r_3$. By cyclic permutation of the coordinate basis, this reproduces the complete $\mathfrak{so}(3)$-Poisson bracket algebra

$$
\boxed{\{r_i, r_j\} = \left\langle \frac{1}{i\hbar}[\sigma_i, \sigma_j] \right\rangle = \frac{2}{\hbar}\epsilon_{ijk} r_k.} \qquad (3.56)
$$

Hamiltonian Action and the Moment Map

The group $G = SU(2)$ acts on $\mathbb{P}\mathcal{H}(\mathbb{C}^2)$ through projective unitary transformations, playing the role that $SO(3)$ played for the classical sphere of Section **??**. Its Lie algebra $\mathfrak{su}(2)$ is spanned by the skew-adjoint generators $\xi_j := -\frac{i}{2}\sigma_j$, normalized so that $i\hbar\xi_j = \frac{\hbar}{2}\sigma_j$ is the physical spin operator. By Proposition 3.6.1, the induced action is strongly Hamiltonian, and its momentum map $\mu : \mathbb{P}\mathcal{H}(\mathbb{C}^2) \to \mathfrak{su}(2)^*$ is given pointwise by

$$
\langle \mu([\psi]), \xi_j \rangle = \langle i\hbar\xi_j \rangle([\psi]) = \frac{\hbar}{2} r_j([\psi]).
$$

Identifying $\mathfrak{su}(2)^* \cong \mathbb{R}^3$ through the three components $\langle \mu, \xi_j \rangle$, $j = 1, 2, 3$, this yields

$$
\boxed{\mu = \frac{\hbar}{2}\mathbf{r}.} \qquad (3.57)
$$

The momentum map thus corresponds to the quantum spin angular momentum vector, the $\hbar/2$-rescaled counterpart of the classical moment map $\mu_{\mathrm{cl}}(\mathbf{r}) = -\mathbf{r}$ of Equation (**??**).

Noether's Theorem

The momentum map constructed above is the geometric counterpart of a conserved quantity associated with a continuous symmetry, in the sense of the Quantum Noether Theorem 3.11. We illustrate this correspondence through the precession of a spin placed in a uniform magnetic field, governed by the same Kähler structure fixed above.

Consider the Hamiltonian $H = \frac{\hbar\omega}{2}\sigma_3$, describing a spin in a uniform magnetic field along the z-axis and invariant under rotations about it. The symmetry generator is $\xi_3 = -\frac{i}{2}\sigma_3$. The associated Noether invariant is

$$
\langle \mu, \xi_3 \rangle = \frac{\hbar}{2} r_3.
$$

As computed in Equation (3.57), and since $[H, i\hbar\xi_3] = \left[\frac{\hbar\omega}{2}\sigma_3, \frac{\hbar}{2}\sigma_3\right] = 0$, the Quantum Noether Theorem 3.11 guarantees that r_3 is conserved. In fact, writing $h := \langle H \rangle = \frac{\hbar\omega}{2} r_3$, Equation (3.56) together with $\dot{f} = \{f, h\}$, descending from the proof of Proposition 3.3.4, yields the equations of motion for the Bloch vector along the Hamiltonian flow of h:

$$
\begin{aligned}
\dot{r}_1 &= \{r_1, h\} = -\omega r_2,\\
\dot{r}_2 &= \{r_2, h\} = \omega r_1,\\
\dot{r}_3 &= \{r_3, h\} = 0.
\end{aligned}
$$

Solving yields a uniform precession about the z-axis:

$$
r_1(t) + i r_2(t) = e^{i\omega t}\big(r_1(0) + i r_2(0)\big), \qquad r_3(t) = r_3(0).
$$

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